

HR / POWER

Report di identificazione

Run afb32cd6fecb4d82bc0b4d0de1d47294

Dataset: G1_UtD

Creazione: 2026-09-28T18:35:01.928493+00:00 | Stato: completed

Struttura: p1d_full | $\tau dx/dt + x = K (P(t-L)-P_0)$; $HR_{hat} = B+x$

Inizializzazione: estimated_equilibrium | $P_0 = \text{mean}(P_{pre})$; $HR_0 = HR(0)$; $x(0) = HR_0 - B$; B estimated; $P(t < 0) = P_0$

Dataset: 56 segmenti; train richiesti: 34; risultati: 34; validi: 34; falliti: 0.

Ambito: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

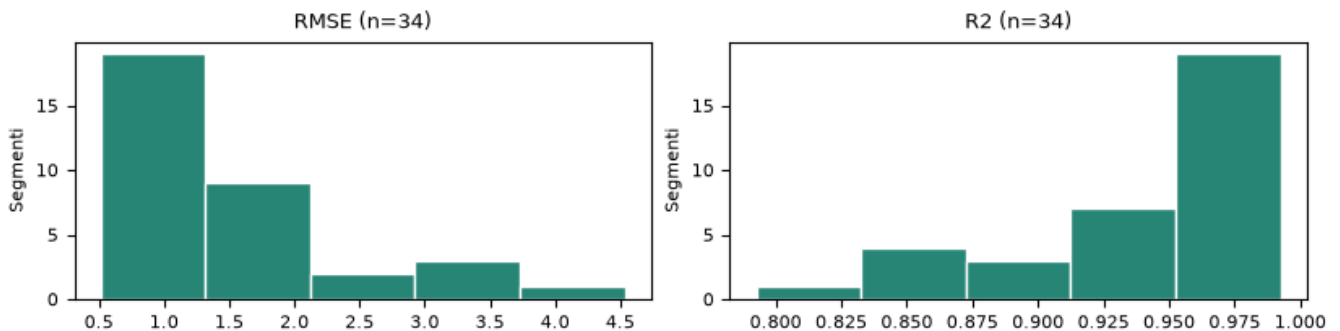
Pre-window richiesta: si; durata: 10 s. P0: pre_window_mean; HR0: first_segment_sample

Pre-window disponibili: 34 / 34 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

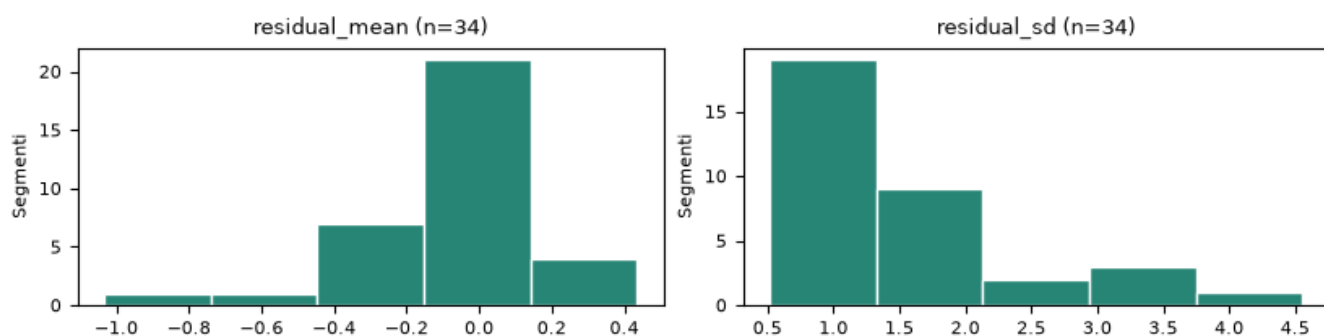
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	34	1.4827	1.2209	0.93328	0.51228	4.5338
MAE	34	1.2188	1.0038	0.80023	0.42722	3.9292
bias	34	-0.077248	-0.0057864	0.2712	-1.0337	0.43523
R2	34	0.93722	0.95488	0.047329	0.79256	0.9924
amplitude_ratio	34	1.0159	1.0136	0.069031	0.87568	1.1343



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	34	-0.077248	-0.0057864	0.2712	-1.0337	0.43523
residual_sd	34	1.4773	1.217	0.92499	0.51529	4.5567
residual_autocorrelation_lag1	34	0.79886	0.82175	0.1186	0.45661	0.94051
residual_autocorrelation_lag5	34	0.13136	0.14833	0.21459	-0.36138	0.54423



Segnalazione	Segmenti
at_bounds	18
autocorrelated_residuals	33
equivalent_predictions_different_parameters	3
irregular_sampling	1
multistart_disagreement	33
near_bounds	5
recording_gaps	1

Multistart

Ricerca: profile_multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 544; convergenti: 540. Accordo di tutti gli start: 1 / 34 fit validi.

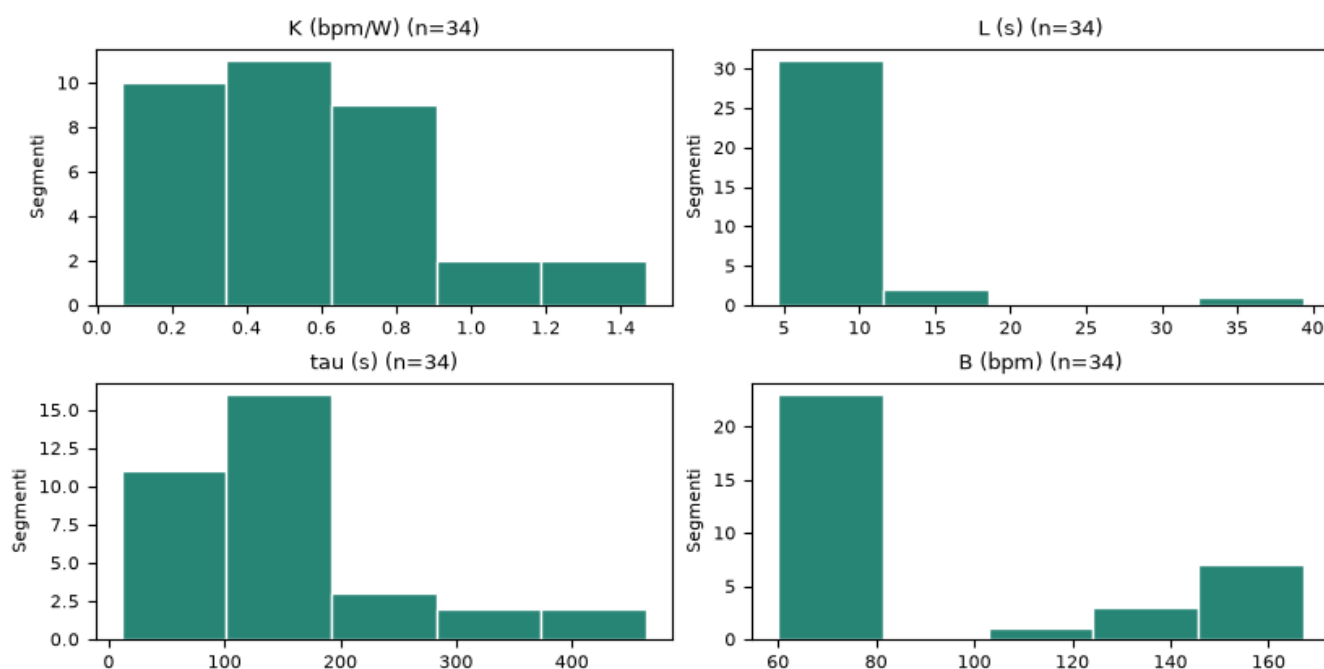
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	544	540.41	96.996	1603.9	11.809	15445
nfev	544	16.561	12	42.689	3	500

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
K (bpm/W)	34	0.54209	0.56666	0.34177	0.065657	1.4687
L (s)	34	8.6834	7.5364	5.7605	4.6006	39.367
tau (s)	34	149.35	136.91	109.47	10.926	464.42
B (bpm)	34	87.436	60	41.124	60	167.02

Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
K	0.61227	88.702	0	0
L	0.52182	6.0255	0	0
tau	144.52	89.677	0	0
B	96.412	160.69	18	23



Identificabilita

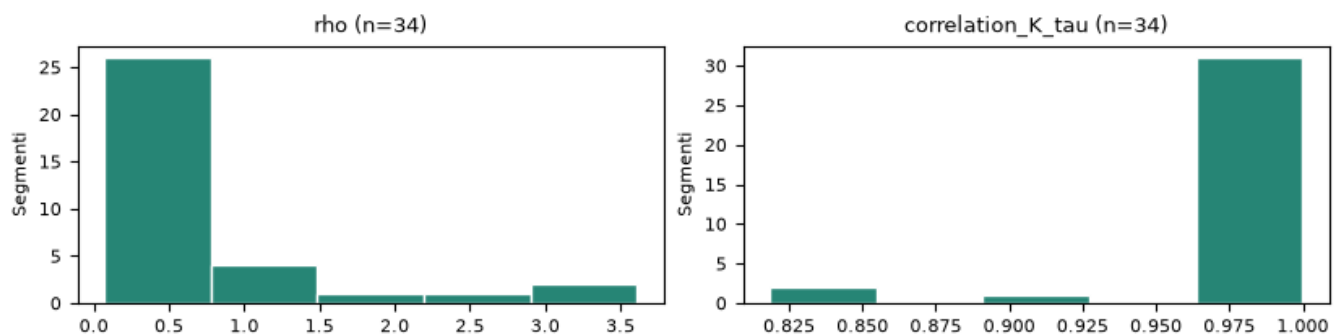
- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unita di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.

- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualita della traiettoria.

- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.

- Multistart: SSE simili con parametri differenti indicano ambiguita. CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	34	0.71235	0.3309	0.88718	0.067748	3.6106
correlation_K_tau	34	0.98462	0.99925	0.042064	0.8183	0.99995
Jacobian rank	34	4	4	0	4	4
Jacobian condition	34	41638	45686	27968	806.59	99156



Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_022-023.csv

ID: 00412a3a52332d7ce2dc0d21 | Stato: fitted

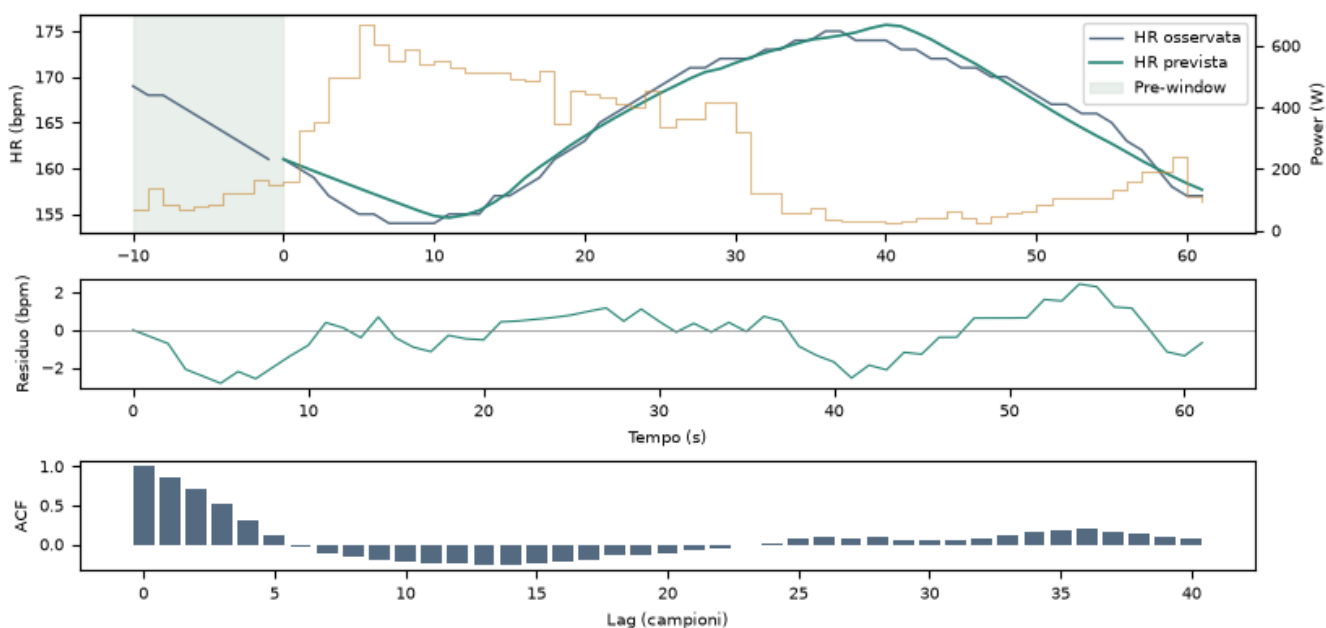
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=108.5 W; HR0=161 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=101 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2374	0.99663	-0.24998	0.96891	1.0044

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.24998	1.2217	0.85707	0.11291



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.61421	0.66374	-0.68673 / 1.9152	108.06	no	no
L	9.6925	0.58413	8.5476 / 10.837	6.0266	no	no
tau	156.51	169.64	-175.98 / 489	108.39	no	no
B	60	113.78	-163.01 / 283.01	189.64	si	si

rho=0.32782; Jacobian rank=4/4; cond=52673; accordo multistart=no.

Correlazioni: {"K_L": -0.9506973654540947, "K_tau": 0.9996775044374357, "K_B": -0.9999585781692797, "L_tau": -0.9521475953364921, "L_B": 0.9508650896544035, "tau_B": -0.9995058279478921}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 94.925 / 2237

Griglia: 42441 coppie L/tau; SSE multistart originale: 94.925; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_074-075.csv

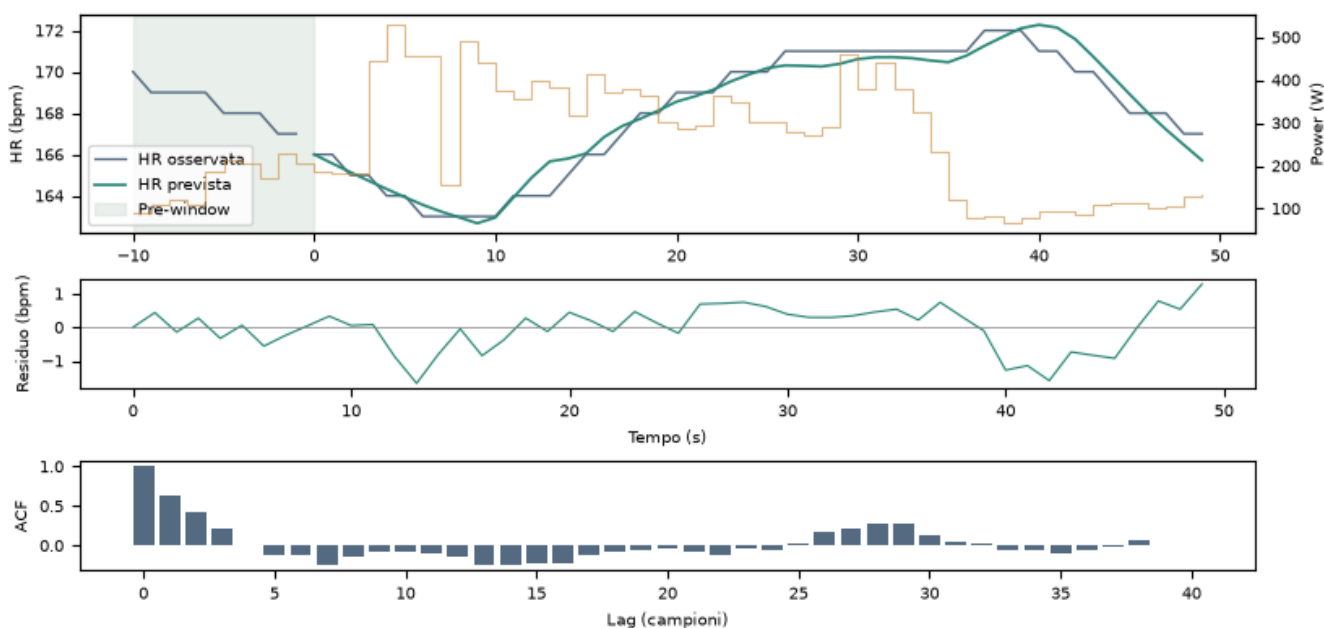
ID: 0898bb8abd4b4f4dd42982da | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=164.4 W; HR0=166 bpm.

HR(0) osservata fissata; B=149.79 bpm stimato; x(0)=16.209 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.63422	0.49236	-0.024287	0.95429	1.067
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.024287	0.64019	0.62814	-0.12912	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.1523	0.033006	0.087611 / 0.217	21.671	no	no
L	6.4882	0.27568	5.9478 / 7.0285	4.249	no	no
tau	36.766	8.4646	20.176 / 53.357	23.023	no	no
B	149.79	3.9021	142.14 / 157.44	2.605	no	no

rho=1.1563; Jacobian rank=4/4; cond=7180.7; accordo multistart=no.

Correlazioni: {"K_L": -0.8020490902787116, "K_tau": 0.9796378773900383, "K_B": -0.9977764790557746, "L_tau": -0.8135038749821687, "L_B": 0.80010056760771, "tau_B": -0.9685058092190437}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 20.112 / 246.78

Griglia: 42441 coppie L/tau; SSE multistart originale: 20.112; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_014-015.csv

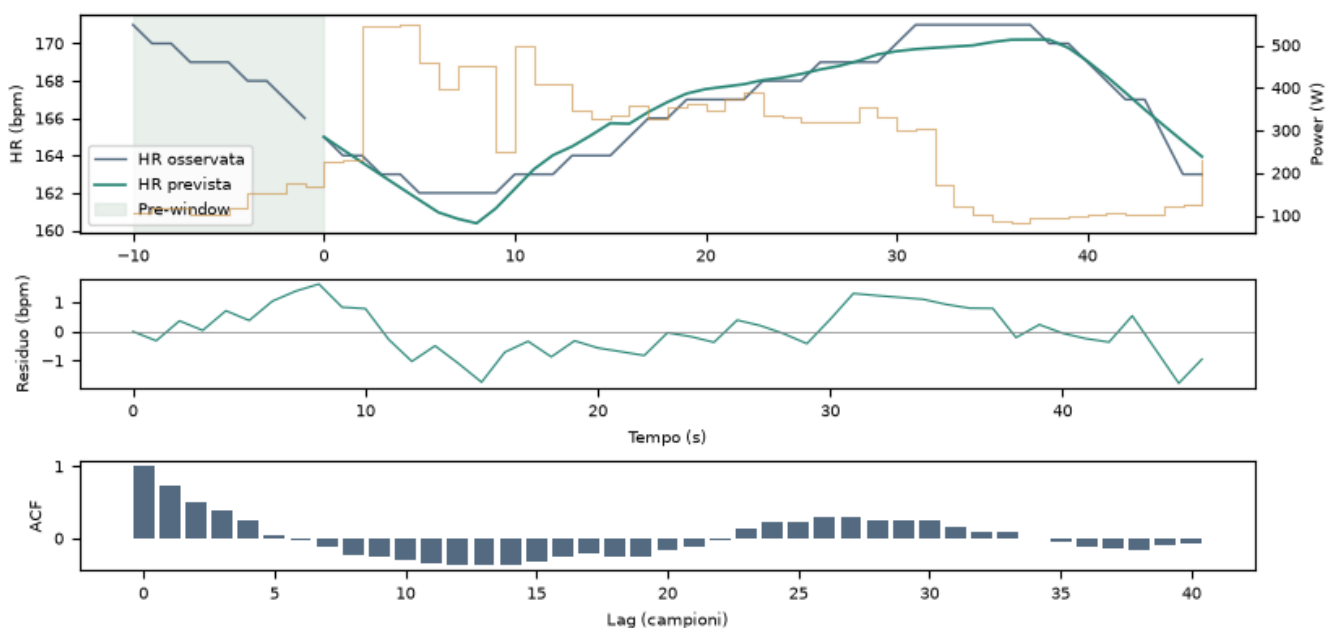
ID: 0b4df52239328d0be144febd | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=130.7 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.79699	0.65339	0.037836	0.93302	1.0925
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.037836	0.8047	0.7229	0.040406	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.63238	0.91395	-1.159 / 2.4237	144.53	no	no
L	6.2007	0.38447	5.4471 / 6.9543	6.2005	no	no
tau	152.93	222.44	-283.05 / 588.91	145.45	no	no
B	60	153.68	-241.22 / 361.22	256.14	no	si

rho=0.26025; Jacobian rank=4/4; cond=66285; accordo multistart=no.

Correlazioni: {"K_L": -0.8894136855603829, "K_tau": 0.9993848496087439, "K_B": -0.9999726808385296, "L_tau": -0.8892468944966927, "L_B": 0.8906874236682611, "tau_B": -0.9991993763703687}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 29.854 / 302.48

Griglia: 42441 coppie L/tau; SSE multistart originale: 29.854; selezione: multistart.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_028-029.csv

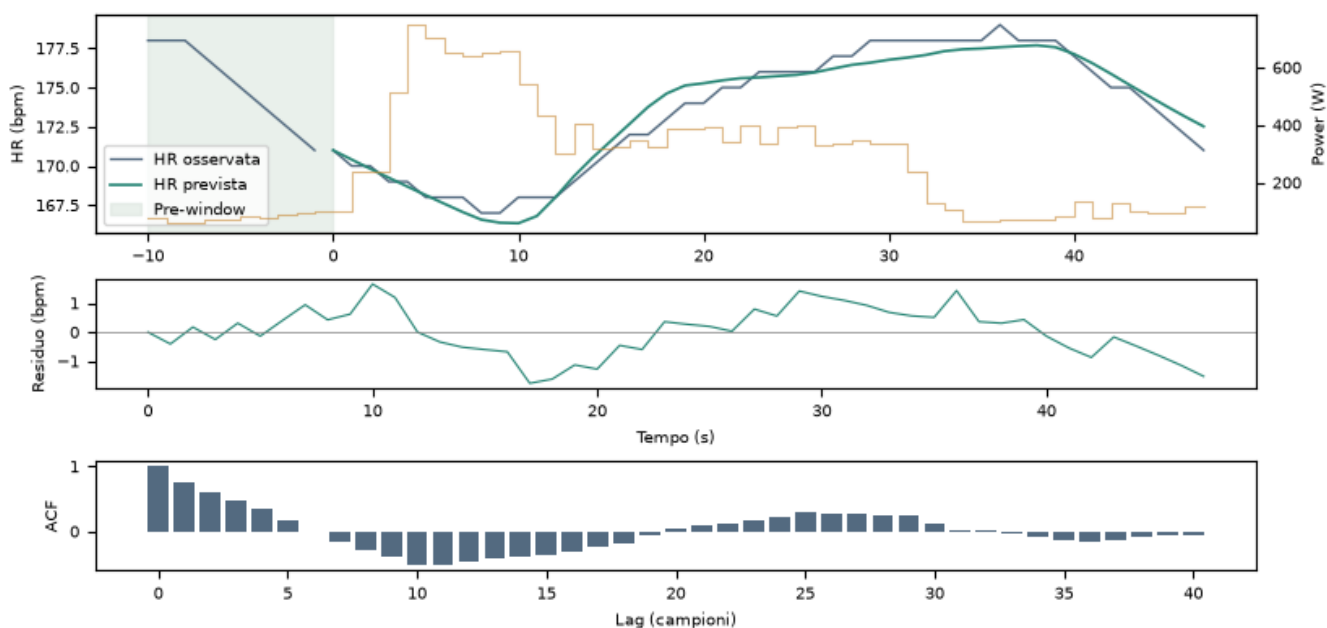
ID: 101b42598bd2beff0cbfd3e6 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=78.7 W; HR0=171 bpm.

HR(0) osservata fissata; B=135.99 bpm stimato; x(0)=35.008 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.82073	0.67276	0.03001	0.9538	0.94558
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.03001	0.82886	0.7594	0.18258	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.17405	0.075484	0.0261 / 0.322	43.369	no	no
L	7.3798	0.38504	6.6251 / 8.1345	5.2175	no	no
tau	58.602	25.824	7.9862 / 109.22	44.067	no	no
B	135.99	16.187	104.27 / 167.72	11.903	no	no

rho=0.67609; Jacobian rank=4/4; cond=22667; accordo multistart=no.

Correlazioni: {"K_L": -0.8790353172606559, "K_tau": 0.9944860520202572, "K_B": -0.9994396675617272, "L_tau": -0.8756507947899206, "L_B": 0.8829132102205652, "tau_B": -0.9914715559950233}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 32.333 / 394.56

Griglia: 42441 coppie L/tau; SSE multistart originale: 32.333; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_066-067.csv

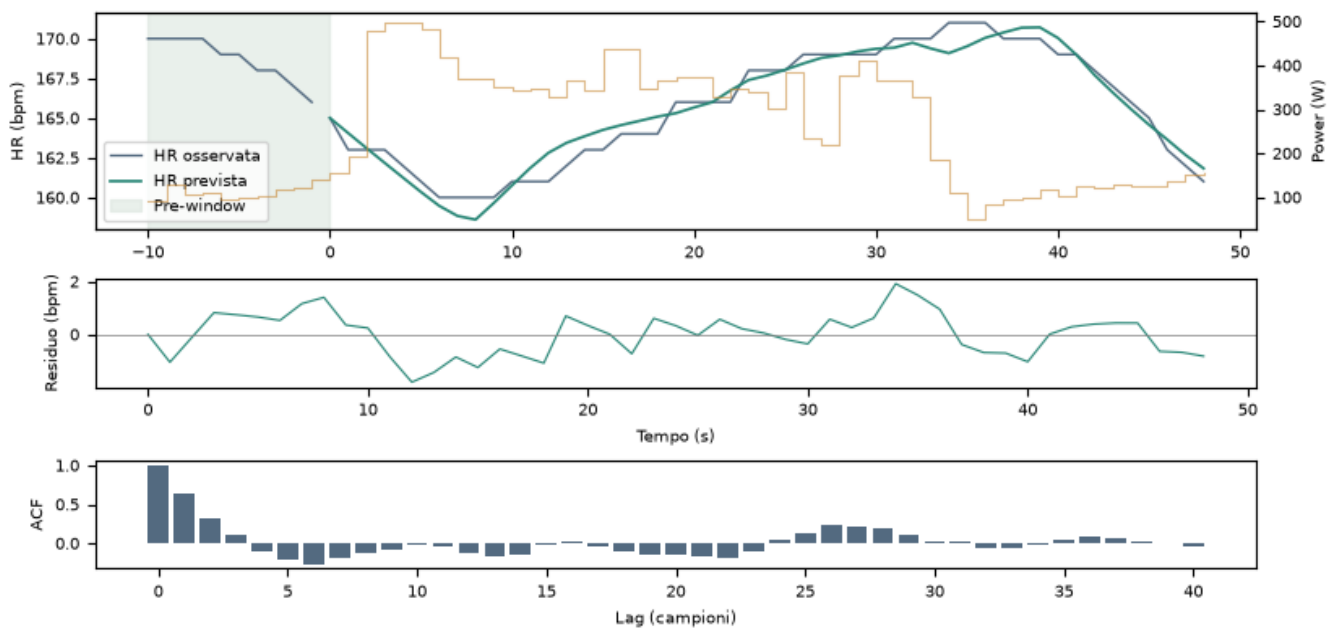
ID: 12c32d6f14eb454003daf51e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=111 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.79686	0.65629	-0.00095915	0.95056	1.1007
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.00095915	0.80511	0.63597	-0.219	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.57702	0.49304	-0.38934 / 1.5434	85.446	no	no
L	5.8476	0.28958	5.28 / 6.4152	4.952	no	no
tau	110.2	94.321	-74.668 / 295.07	85.59	no	no
B	60	90.338	-117.06 / 237.06	150.56	si	si

rho=0.3825; Jacobian rank=4/4; cond=46627; accordo multistart=no.

Correlazioni: {"K_L": -0.8825685584881103, "K_tau": 0.9989883712092557, "K_B": -0.9999693979535124, "L_tau": -0.8793224194490339, "L_B": 0.8843049284361488, "tau_B": -0.9987978174913155}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 31.114 / 519.53

Griglia: 42441 coppie L/tau; SSE multistart originale: 31.114; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_026-027.csv

ID: 1822b3e6cfec7ec77aa201 | Stato: fitted

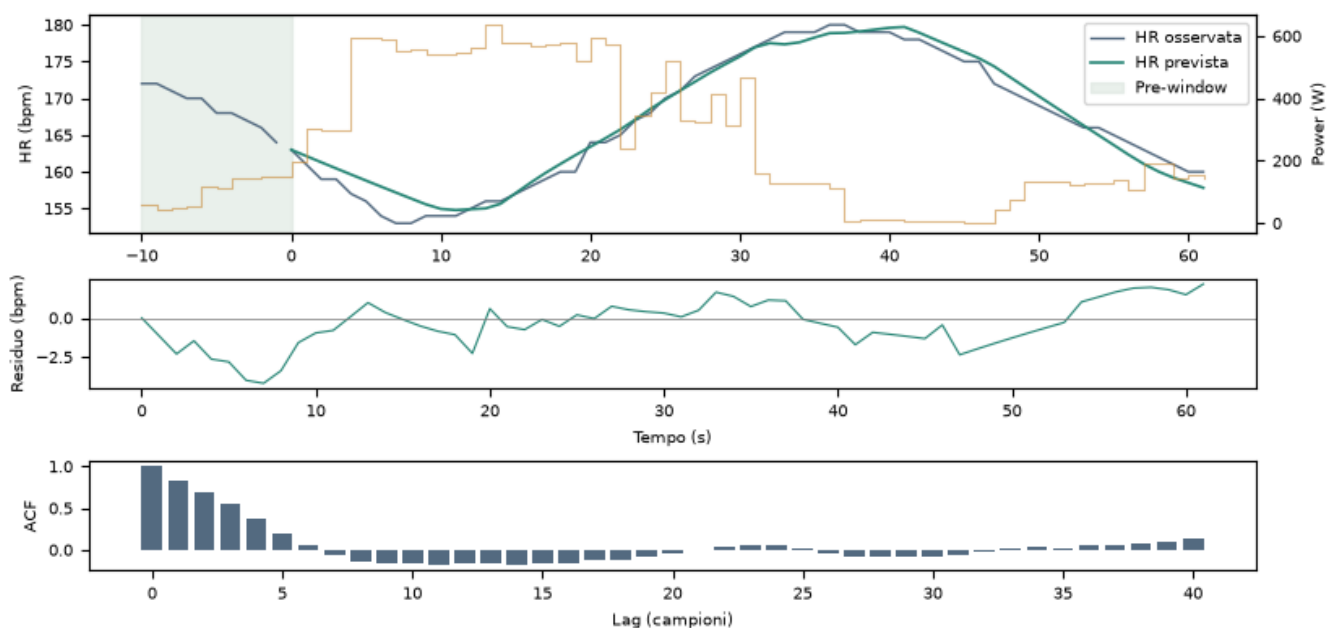
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=102.7 W; HR0=163 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=103 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.4945	1.1753	-0.39637	0.97048	0.92276

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.39637	1.4527	0.82141	0.19879



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.5504	0.46539	-0.36177 / 1.4626	84.555	no	no
L	9.6086	0.57886	8.474 / 10.743	6.0244	no	no
tau	120.33	102.42	-80.415 / 321.08	85.116	no	no
B	60	90.744	-117.86 / 237.86	151.24	si	si

rho=0.42707; Jacobian rank=4/4; cond=40057; accordo multistart=no.

Correlazioni: {"K_L": -0.9539487223581111, "K_tau": 0.9995212166852065, "K_B": -0.9999457427333427, "L_tau": -0.9553071072245846, "L_B": 0.9542843592394364, "tau_B": -0.9993120166287358}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 138.47 / 3629.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 138.47; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_024-025.csv

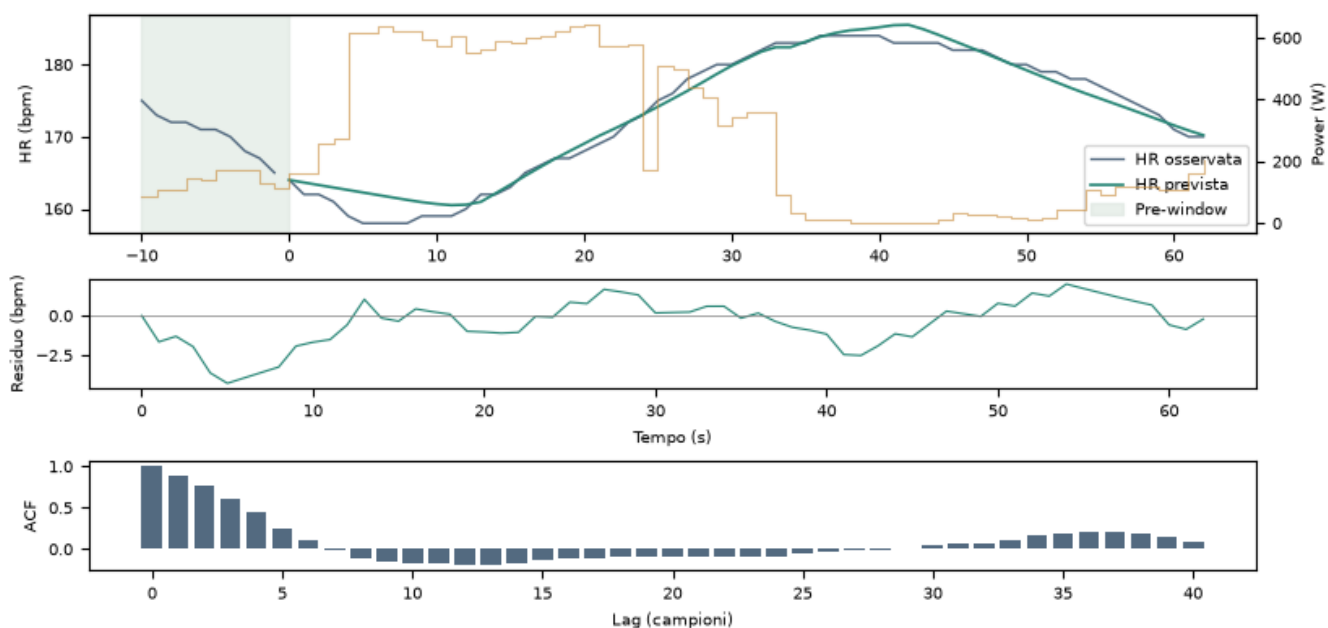
ID: 1ab958790772ba8ceb12c0ab | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=135 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.4991	1.1285	-0.42444	0.97223	0.96165
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.42444	1.4493	0.8873	0.25589	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.9381	1.4356	-1.8757 / 3.7519	153.04	no	no
L	8.7273	0.63879	7.4752 / 9.9793	7.3195	no	no
tau	297.41	453.56	-591.57 / 1186.4	152.5	no	no
B	60	172.53	-278.15 / 398.15	287.54	si	si

rho=0.17912; Jacobian rank=4/4; cond=62361; accordo multistart=no.

Correlazioni: {"K_L": -0.9325027452084493, "K_tau": 0.9997662589109708, "K_B": -0.9998028563462217, "L_tau": -0.9342886478275598, "L_B": 0.931440561967018, "tau_B": -0.9992022514500025}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 141.58 / 2289.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 141.58; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260407082440__laps_076-077.csv

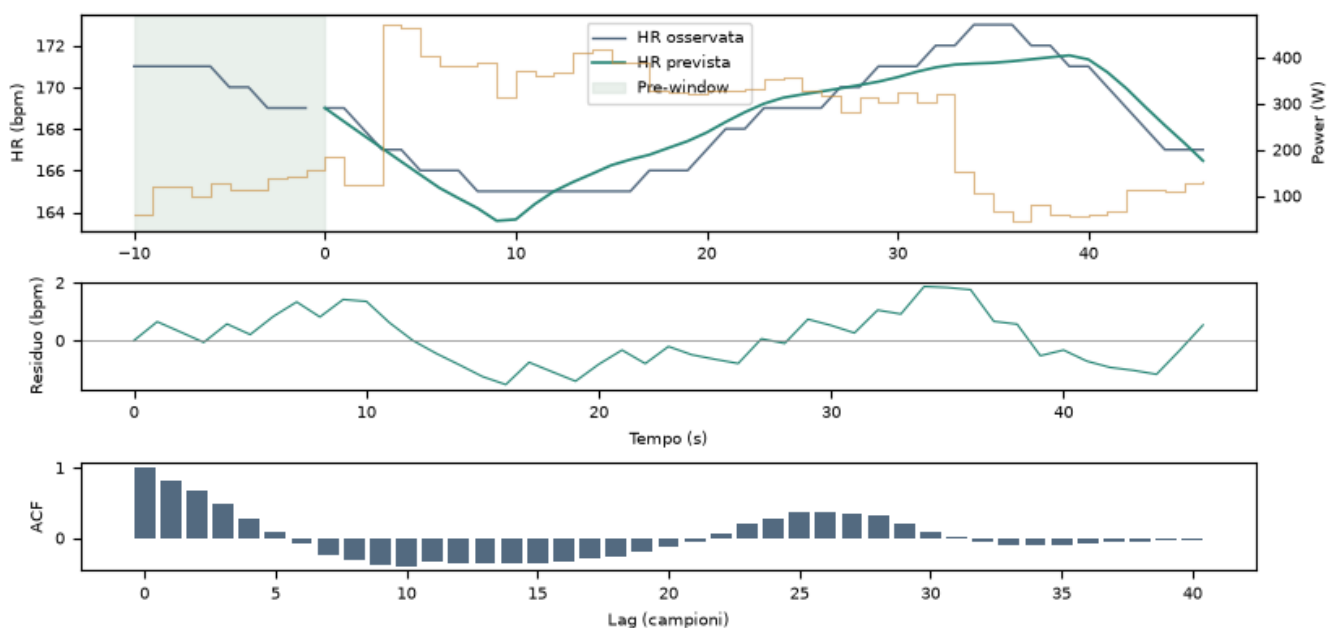
ID: 1cfb031ad5d32abfb3381a2a | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=118.7 W; HR0=169 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=109 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.89436	0.75408	0.046907	0.87866	0.99287
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.046907	0.90279	0.81914		0.07928



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.65525	1.6473	-2.5735 / 3.884	251.4	no	no
L	6.5117	0.53122	5.4705 / 7.5529	8.1579	no	no
tau	167.53	423.62	-662.77 / 997.83	252.86	no	no
B	60	271.85	-472.82 / 592.82	453.08	si	si

rho=0.2357; Jacobian rank=4/4; cond=90310; accordo multistart=no.

Correlazioni: {"K_L": -0.8879236591550242, "K_tau": 0.9997364433230816, "K_B": -0.9999903626413498, "L_tau": -0.884719629040752, "L_B": 0.8891236796249953, "tau_B": -0.9996992034820685}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 37.594 / 331.68

Griglia: 42441 coppie L/tau; SSE multistart originale: 37.594; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_021-023.csv

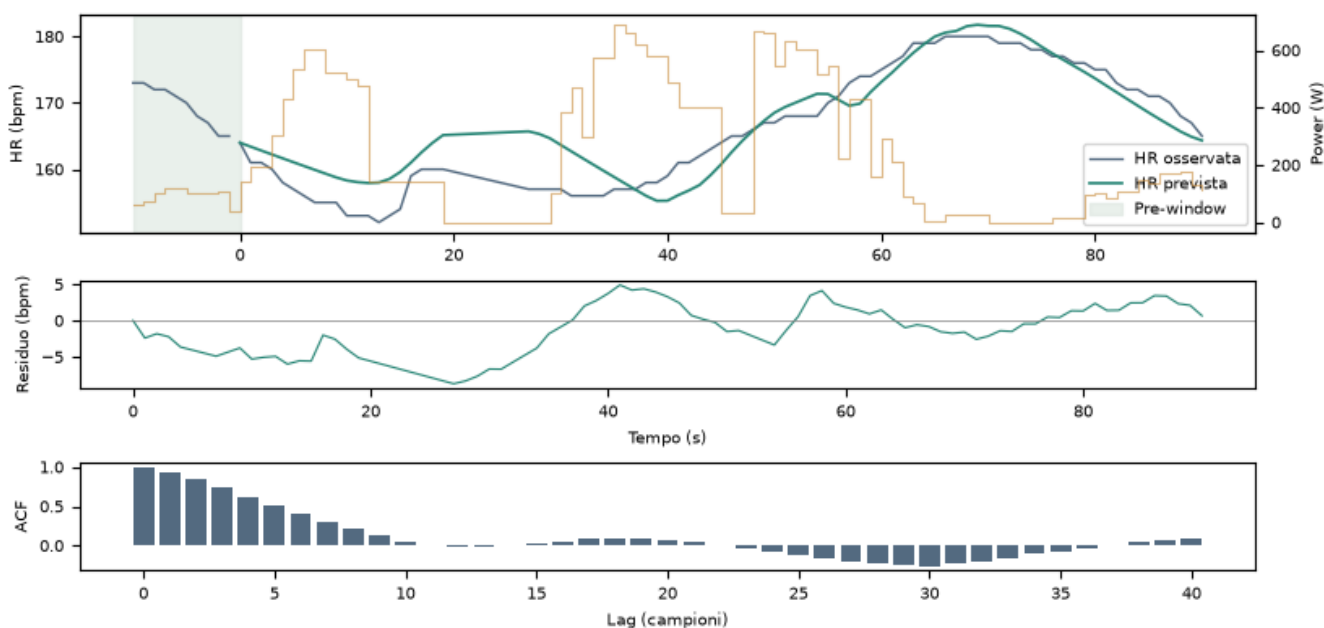
ID: 239cf3ea29845ae82a992d37 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=92.9 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.438	2.8053	-1.0337	0.85207	0.94684
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-1.0337	3.2986	0.94051	0.51886	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.73084	0.62914	-0.50227 / 1.9639	86.084	no	no
L	9.5541	0.51859	8.5377 / 10.571	5.428	no	no
tau	174.66	153.9	-126.99 / 476.3	88.116	no	no
B	60	90.819	-118.01 / 238.01	151.37	si	si

rho=0.46059; Jacobian rank=4/4; cond=20682; accordo multistart=no.

Correlazioni: {"K_L": -0.7078549755199889, "K_tau": 0.9976971185530081, "K_B": -0.99976571407595, "L_tau": -0.7044599470383692, "L_B": 0.7117174058545168, "tau_B": -0.9965874614437544}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 992.87 / 4687.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 992.87; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, recording_gaps, irregular_sampling, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_068-069.csv

ID: 24774320a7b915b740dde850 | Stato: fitted

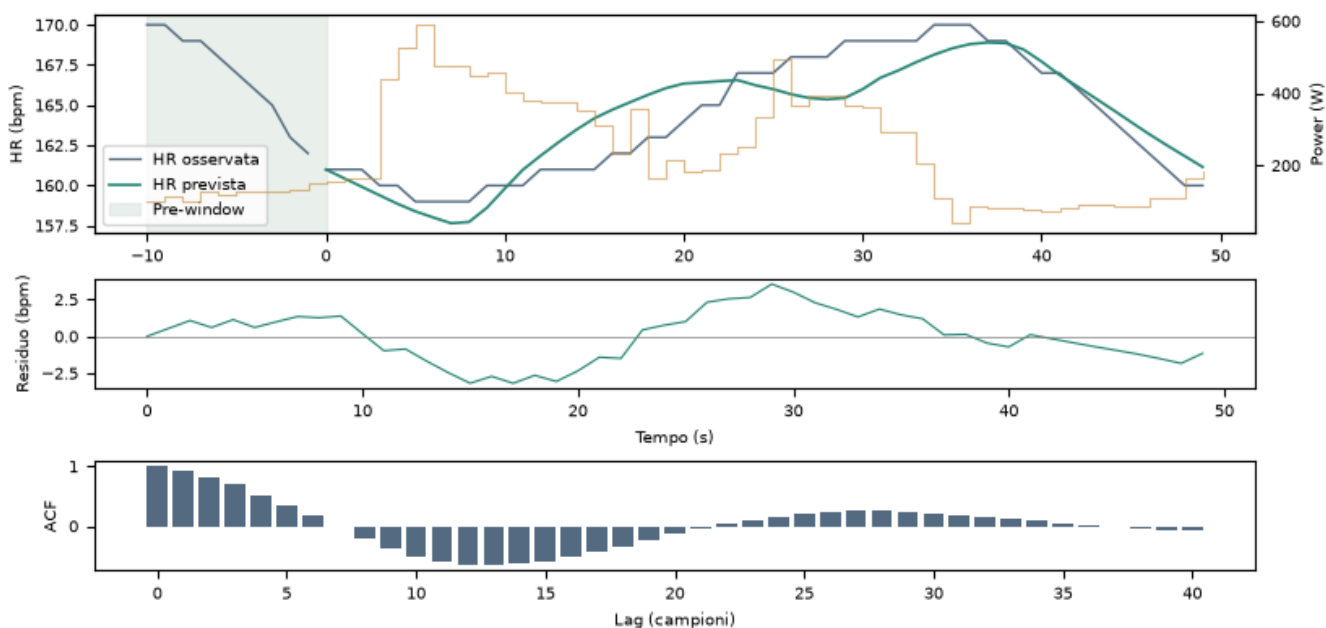
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=123.2 W; HR0=161 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=101 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.7002	1.417	0.011809	0.79256	1.0225

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.011809	1.7174	0.92554	0.3594



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.75406	2.1279	-3.4166 / 4.9248	282.19	no	no
L	4.6006	0.6747	3.2782 / 5.923	14.665	no	no
tau	187.24	525.84	-843.41 / 1217.9	280.84	no	no
B	60	293.98	-516.19 / 636.19	489.96	si	si

rho=0.23713; Jacobian rank=4/4; cond=57052; accordo multistart=no.

Correlazioni: {"K_L": -0.8449403404665354, "K_tau": 0.9993824862285795, "K_B": -0.9999473269998435, "L_tau": -0.8485178853208796, "L_B": 0.8452775656012955, "tau_B": -0.9990555237780401}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 144.53 / 447.55

Griglia: 42441 coppie L/tau; SSE multistart originale: 144.53; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_019-020.csv

ID: 24e7951d1c4d446ab88cdd4e | Stato: fitted

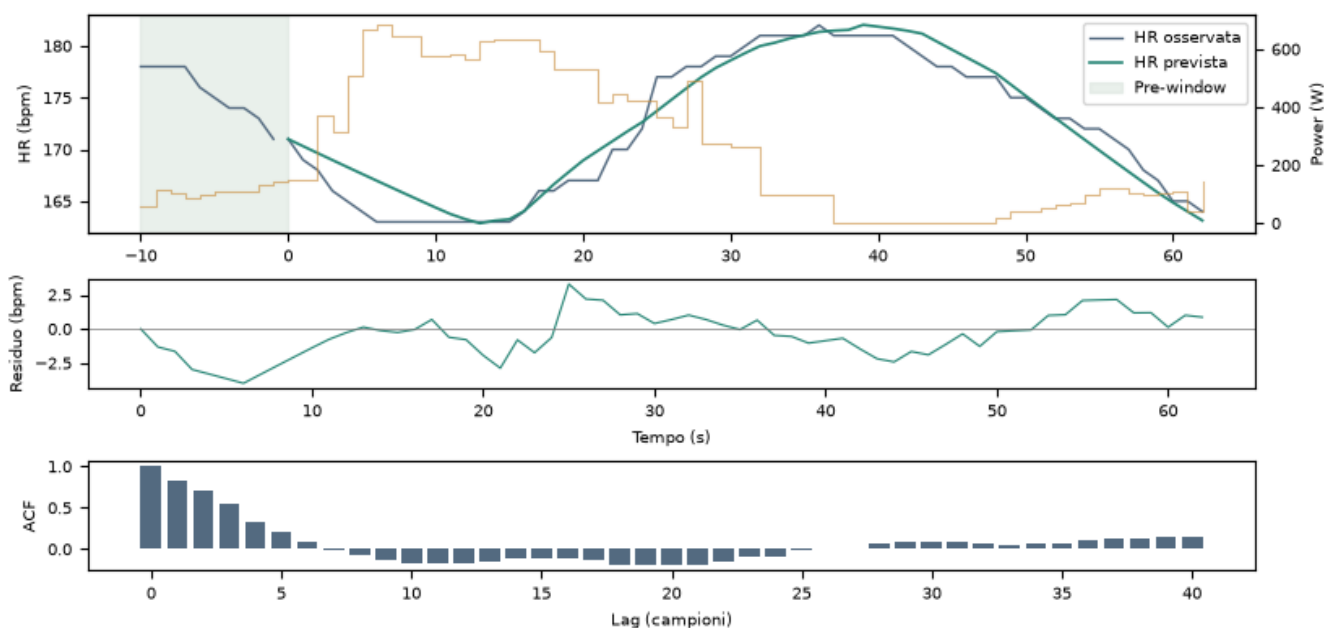
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=107.2 W; HR0=171 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=111 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.623	1.2788	-0.42131	0.94092	1.009

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.42131	1.5799	0.83803	0.20361



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.55462	1.0652	-1.5331 / 2.6423	192.05	no	no
L	10.897	0.89967	9.1341 / 12.661	8.2558	no	no
tau	162.75	314.74	-454.13 / 779.64	193.39	no	no
B	60	216.1	-363.57 / 483.57	360.17	no	si

rho=0.31399; Jacobian rank=4/4; cond=79500; accordo multistart=no.

Correlazioni: {"K_L": -0.965148697659342, "K_tau": 0.9998372541097109, "K_B": -0.9999851203637004, "L_tau": -0.9656909265803799, "L_B": 0.9653799531698898, "tau_B": -0.9997940292580222}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 165.94 / 2520.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 165.94; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_076-077.csv

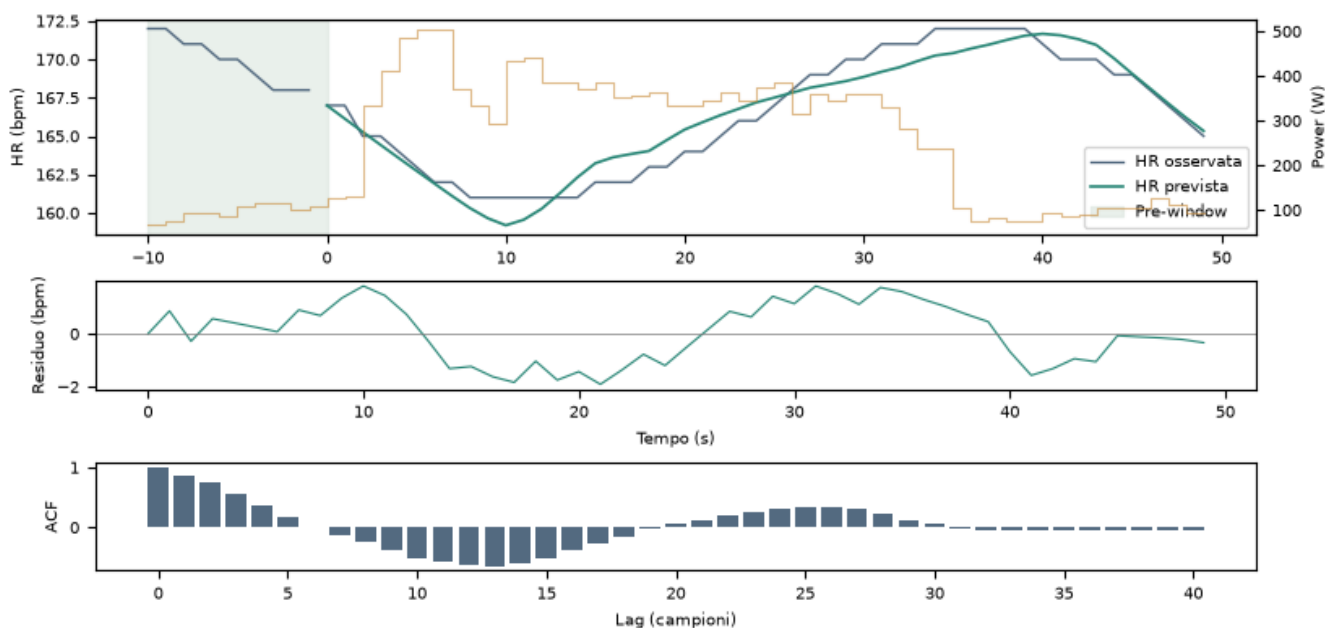
ID: 3d72a3ba651acda8c1791da7 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=96.5 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1036	0.95133	0.033939	0.91685	1.1342
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.033939	1.1143	0.86473	0.17846	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.56629	0.79211	-0.98624 / 2.1188	139.88	no	no
L	7.7967	0.52393	6.7698 / 8.8236	6.7199	no	no
tau	123.37	173.49	-216.68 / 463.42	140.63	no	no
B	60	148.61	-231.28 / 351.28	247.69	no	si

rho=0.33398; Jacobian rank=4/4; cond=55596; accordo multistart=no.

Correlazioni: {"K_L": -0.8942197512824597, "K_tau": 0.999366516350577, "K_B": -0.9999710379828403, "L_tau": -0.8878081016401383, "L_B": 0.8964150482126643, "tau_B": -0.9992141062222683}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 60.895 / 731.95

Griglia: 42441 coppie L/tau; SSE multistart originale: 60.895; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_015-016.csv

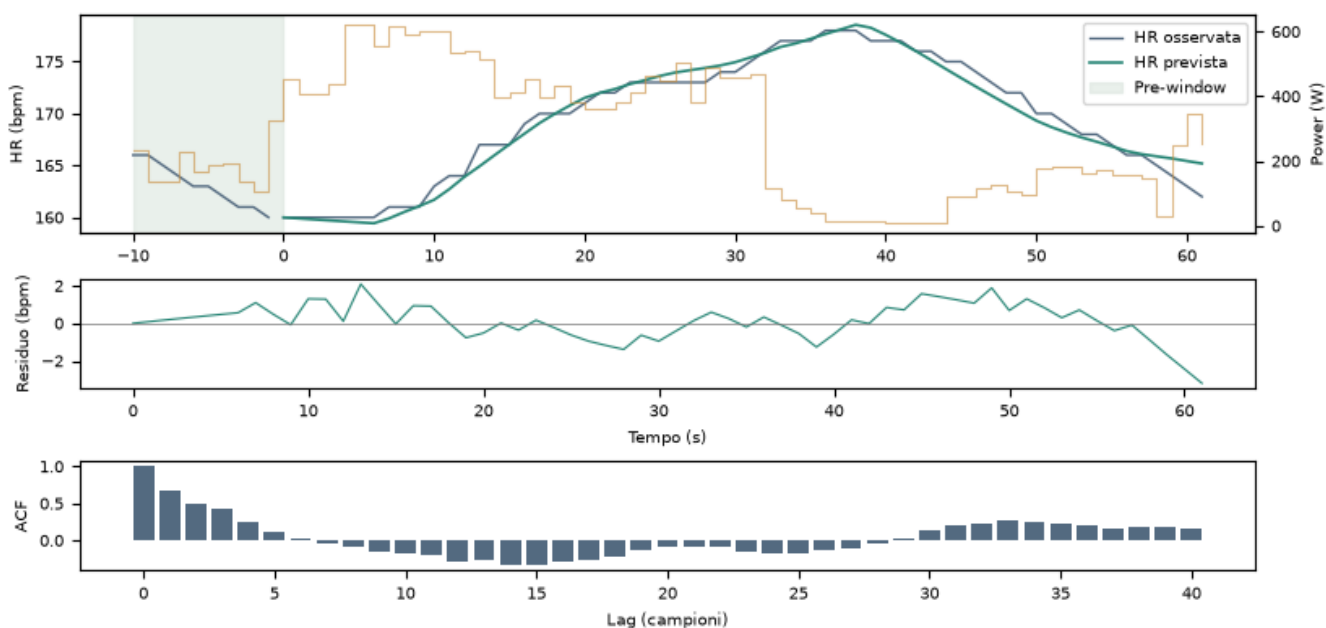
ID: 467e3d3507204e47a1d06103 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=185.5 W; HR0=160 bpm.

HR(0) osservata fissata; B=154.05 bpm stimato; x(0)=5.9525 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.96766	0.72838	0.10111	0.97237	1.0617
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10111	0.97022	0.67883	0.11543	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.18429	0.056125	0.074286 / 0.29429	30.454	no	no
L	6.315	0.47002	5.3937 / 7.2362	7.443	no	no
tau	60.331	17.89	25.266 / 95.395	29.653	no	no
B	154.05	4.8531	144.54 / 163.56	3.1504	no	no

rho=0.90642; Jacobian rank=4/4; cond=10593; accordo multistart=no.

Correlazioni: {"K_L": -0.872757011974183, "K_tau": 0.9961627458392447, "K_B": -0.9948584855138813, "L_tau": -0.8907076028735877, "L_B": 0.8474274061725762, "tau_B": -0.9841287331876654}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 58.055 / 1078.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 58.261; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_062-063.csv

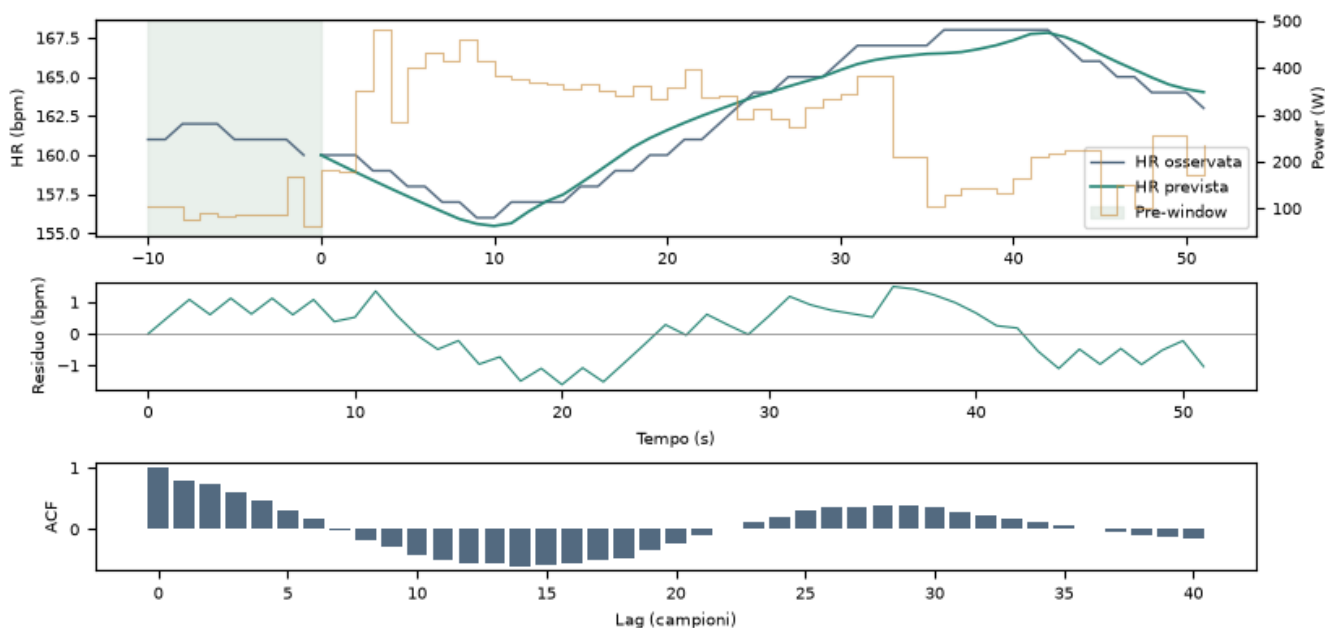
ID: 47eea1ca7bd657074778883b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=95.7 W; HR0=160 bpm.

HR(0) osservata fissata; B=132.99 bpm stimato; x(0)=27.014 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.85059	0.73762	0.096404	0.95547	1.0284
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.096404	0.85336	0.80281	0.31513	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.18771	0.062815	0.064595 / 0.31083	33.463	no	no
L	8.5222	0.52729	7.4887 / 9.5557	6.1873	no	no
tau	48.838	16.809	15.891 / 81.784	34.419	no	no
B	132.99	9.8753	113.63 / 152.34	7.4258	no	no

rho=0.86978; Jacobian rank=4/4; cond=13030; accordo multistart=no.

Correlazioni: {"K_L": -0.8897353839030228, "K_tau": 0.9893632839767252, "K_B": -0.9985270910961273, "L_tau": -0.8672244519223713, "L_B": 0.900627379534499, "tau_B": -0.9818716279938824}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 37.623 / 412.93

Griglia: 42441 coppie L/tau; SSE multistart originale: 37.623; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_017-018.csv

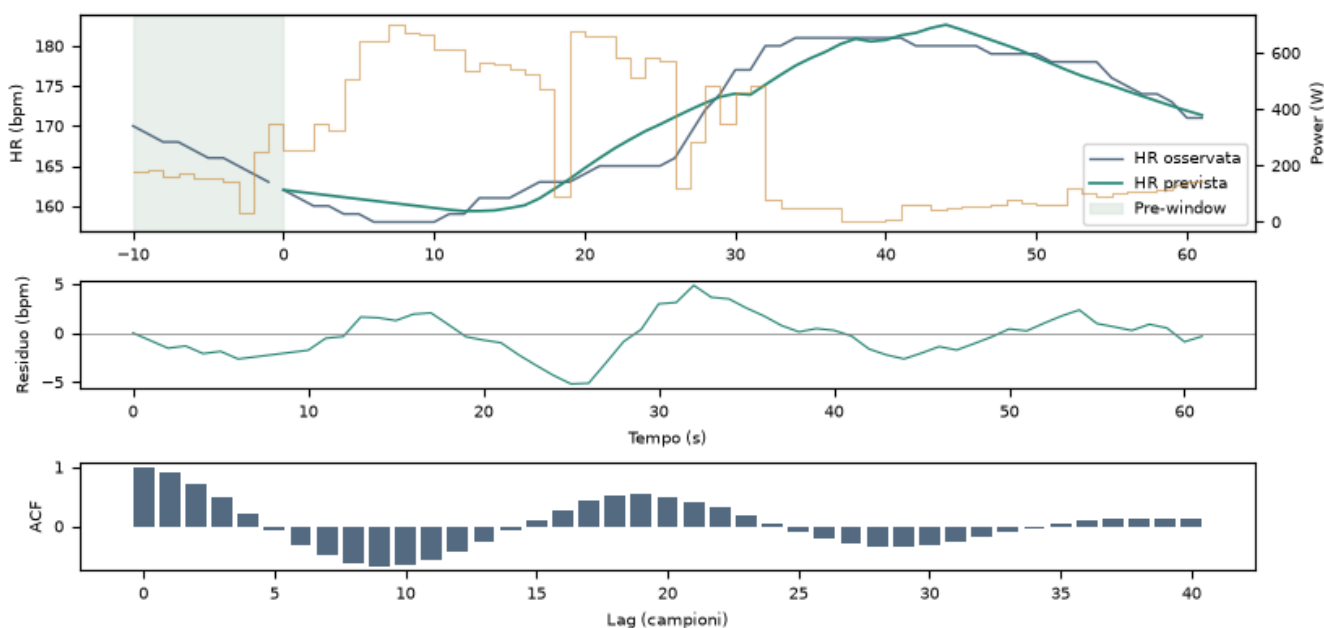
ID: 48e72ea2c9db441feda919d0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=178 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0791	1.6579	-0.28804	0.9421	1.0121
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.28804	2.0759	0.89937	-0.06065	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.3709	3.2094	-4.9195 / 7.6613	234.11	no	no
L	11.789	0.70527	10.407 / 13.172	5.9824	no	no
tau	449.29	1051.5	-1611.6 / 2510.1	234.02	no	no
B	60	256.04	-441.84 / 561.84	426.74	si	si

rho=0.10953; Jacobian rank=4/4; cond=55632; accordo multistart=no.

Correlazioni: {"K_L": -0.8679776499391167, "K_tau": 0.9997059861744098, "K_B": -0.9994211528884323, "L_tau": -0.8691513823334497, "L_B": 0.8673110064980538, "tau_B": -0.998382663009771}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 268.02 / 1722.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 268.02; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuels, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_013-014.csv

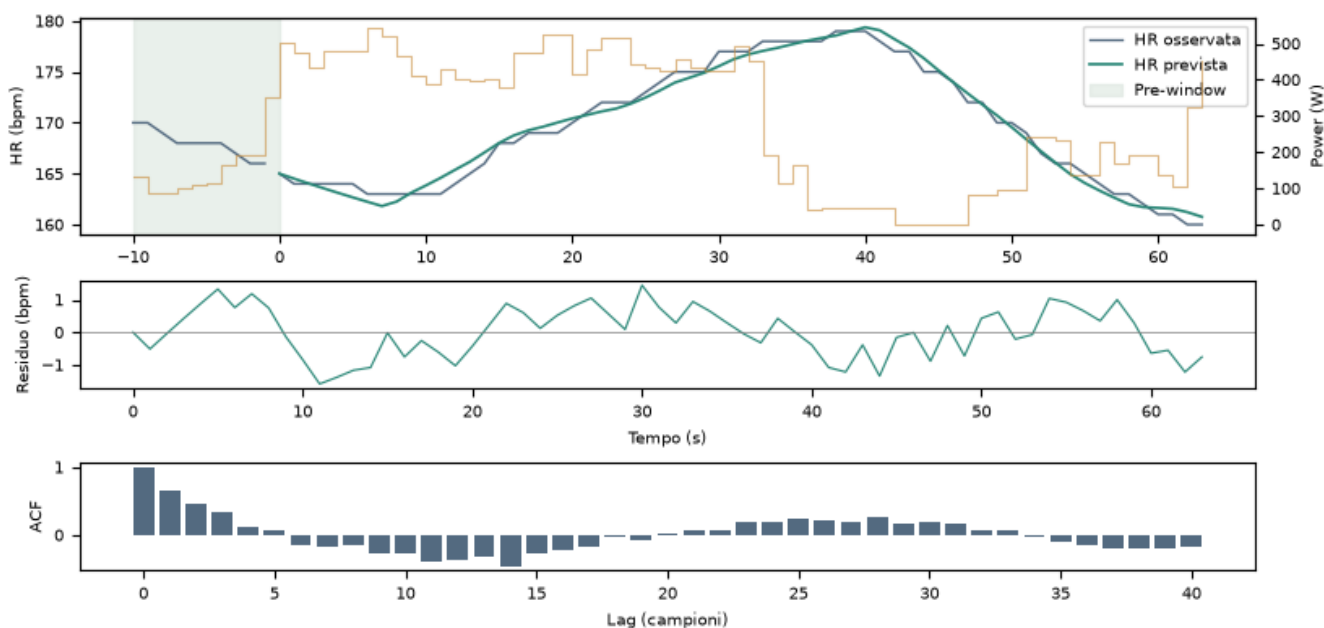
ID: 49069ab3343d75d7654ee1ec | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=153.3 W; HR0=165 bpm.

HR(0) osservata fissata; B=144.47 bpm stimato; x(0)=20.533 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.75583	0.63107	0.011593	0.98385	0.98049
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.011593	0.76172	0.65231	0.058786	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.16405	0.029831	0.10558 / 0.22251	18.184	no	no
L	7.3792	0.31987	6.7522 / 8.0061	4.3347	no	no
tau	41.573	7.9536	25.983 / 57.162	19.132	no	no
B	144.47	4.7102	135.24 / 153.7	3.2604	no	no

rho=1.3379; Jacobian rank=4/4; cond=9694.5; accordo multistart=no.

Correlazioni: {"K_L": -0.8960508548966035, "K_tau": 0.9951778317759562, "K_B": -0.9991422081483201, "L_tau": -0.9117515270986112, "L_B": 0.8922153596425204, "tau_B": -0.9929146302683579}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 36.562 / 1898

Griglia: 42441 coppie L/tau; SSE multistart originale: 37.283; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260501094312__laps_016-017.csv

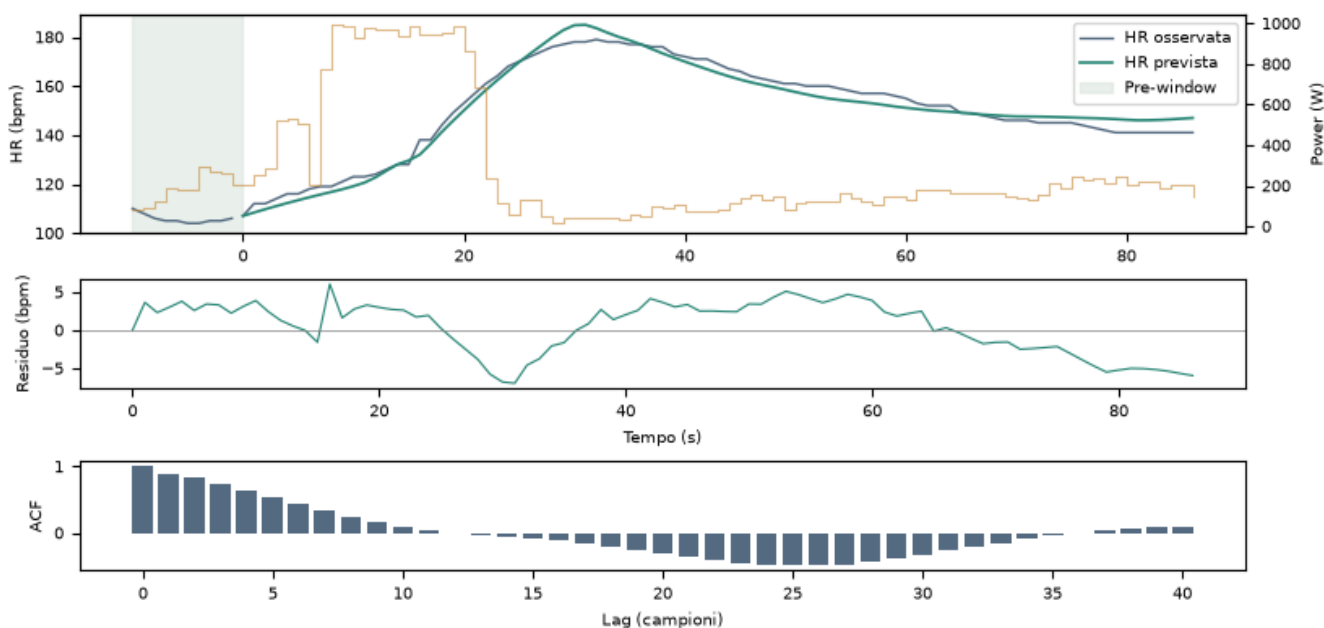
ID: 4e46b2308f9432598cbf72ca | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=186.9 W; HR0=107 bpm.

HR(0) osservata fissata; B=147.19 bpm stimato; x(0)=-40.192 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.4031	2.9976	0.43523	0.96816	1.083
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.43523	3.3947	0.89705	0.54423	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.16204	0.016219	0.13025 / 0.19383	10.009	no	no
L	8.4774	0.38172	7.7292 / 9.2255	4.5028	no	no
tau	28.665	2.61	23.549 / 33.78	9.1053	no	no
B	147.19	0.88602	145.46 / 148.93	0.60195	no	no

rho=2.7045; Jacobian rank=4/4; cond=999.92; accordo multistart=no.

Correlazioni: {"K_L": -0.7613181727413009, "K_tau": 0.9744380281611169, "K_B": -0.7177213486483007, "L_tau": -0.7971393703562756, "L_B": 0.4100483675546141, "tau_B": -0.6136245577410498}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1007.6 / 15445

Griglia: 42441 coppie L/tau; SSE multistart originale: 1007.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_026-027.csv

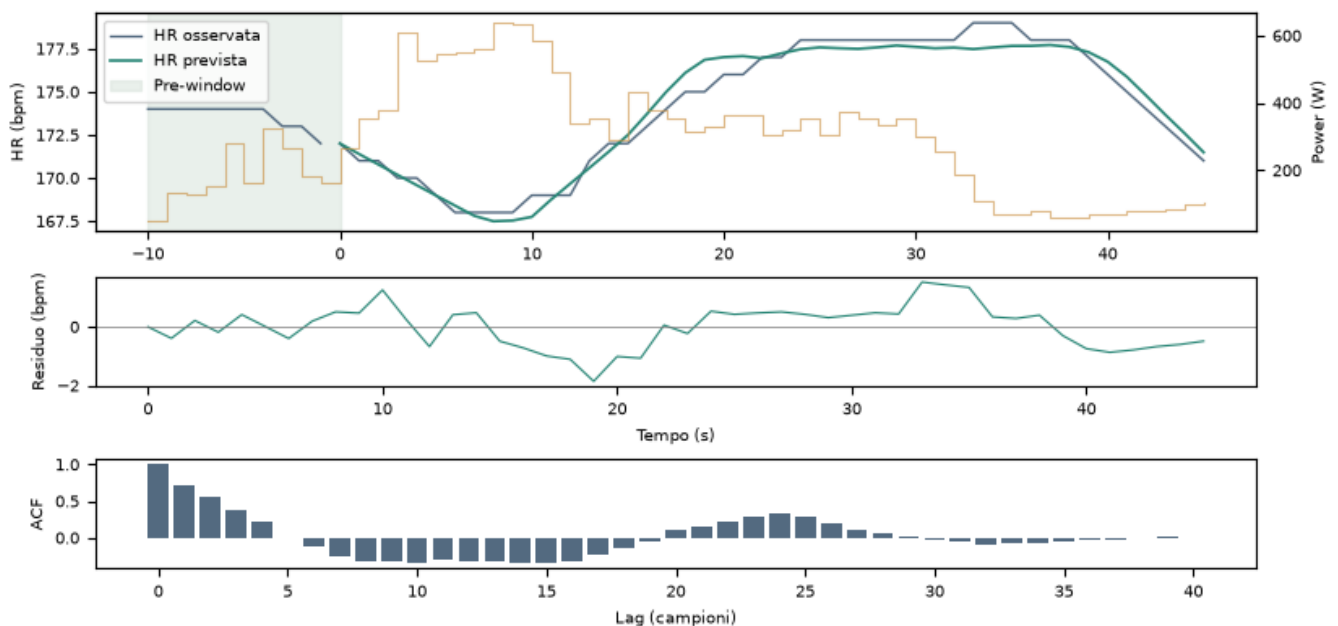
ID: 57b47931dfdd06086ce2e322 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=184.5 W; HR0=172 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=112 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.7157	0.58988	-0.00019838	0.96306	0.92946
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.00019838	0.72361	0.72419		0.0056137



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.77502	1.0437	-1.2707 / 2.8207	134.67	no	no
L	7.2195	0.32634	6.5799 / 7.8591	4.5203	no	no
tau	183.51	248.64	-303.82 / 670.84	135.49	no	no
B	60	153.33	-240.52 / 360.52	255.55	si	si

rho=0.20587; Jacobian rank=4/4; cond=61127; accordo multistart=no.

Correlazioni: {"K_L": -0.8940059571094067, "K_tau": 0.9995907399739852, "K_B": -0.9999664933889776, "L_tau": -0.8949275218784022, "L_B": 0.8948713984312224, "tau_B": -0.9994011824034179}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.562 / 402.55

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.562; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_011-012.csv

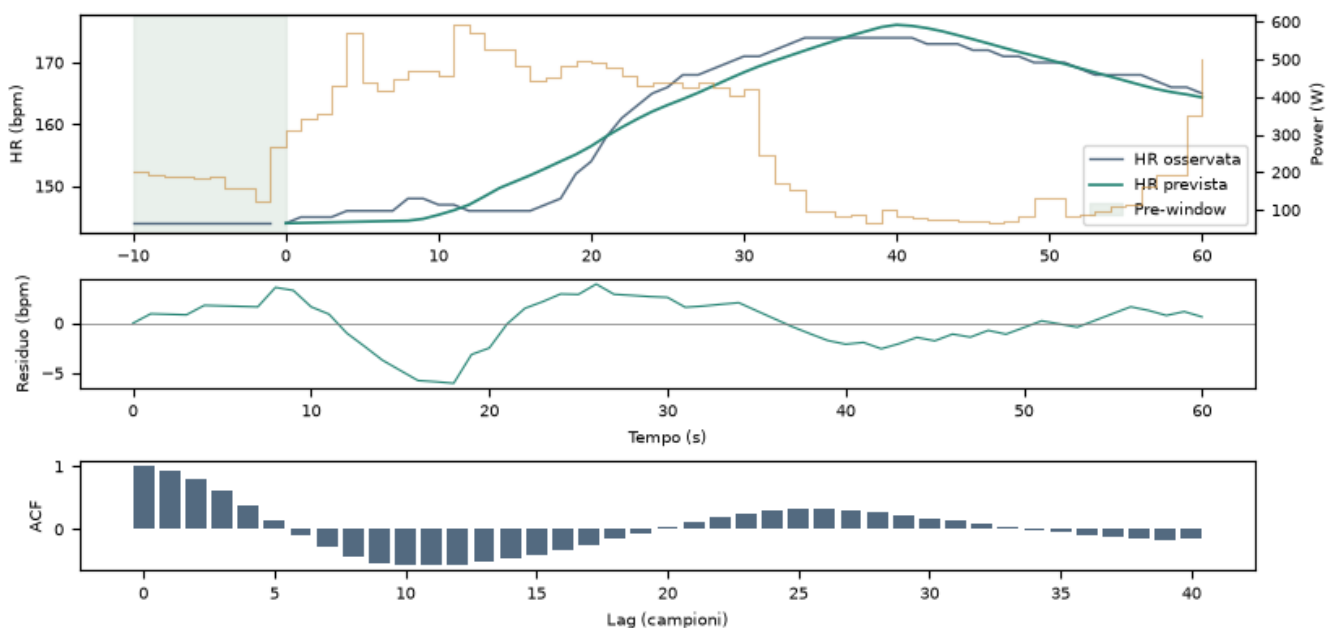
ID: 811df6313aef0591fb62ef23 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=185.1 W; HR0=144 bpm.

HR(0) osservata fissata; B=149.2 bpm stimato; x(0)=-5.2017 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.3206	1.8753	0.064711	0.9582	1.0701
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.064711	2.3389	0.92603	0.14531	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.40555	0.25236	-0.089085 / 0.90018	62.228	no	no
L	8.5292	1.0946	6.3839 / 10.675	12.833	no	no
tau	96.092	58.564	-18.694 / 210.88	60.946	no	no
B	149.2	8.0403	133.44 / 164.96	5.3888	no	no

rho=0.53564; Jacobian rank=4/4; cond=9252.9; accordo multistart=no.

Correlazioni: {"K_L": -0.9197151799184659, "K_tau": 0.9956431710924786, "K_B": -0.9152642617439868, "L_tau": -0.9328210129629863, "L_B": 0.775976288674427, "tau_B": -0.8758015612210681}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 328.49 / 1934.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 328.49; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_070-071.csv

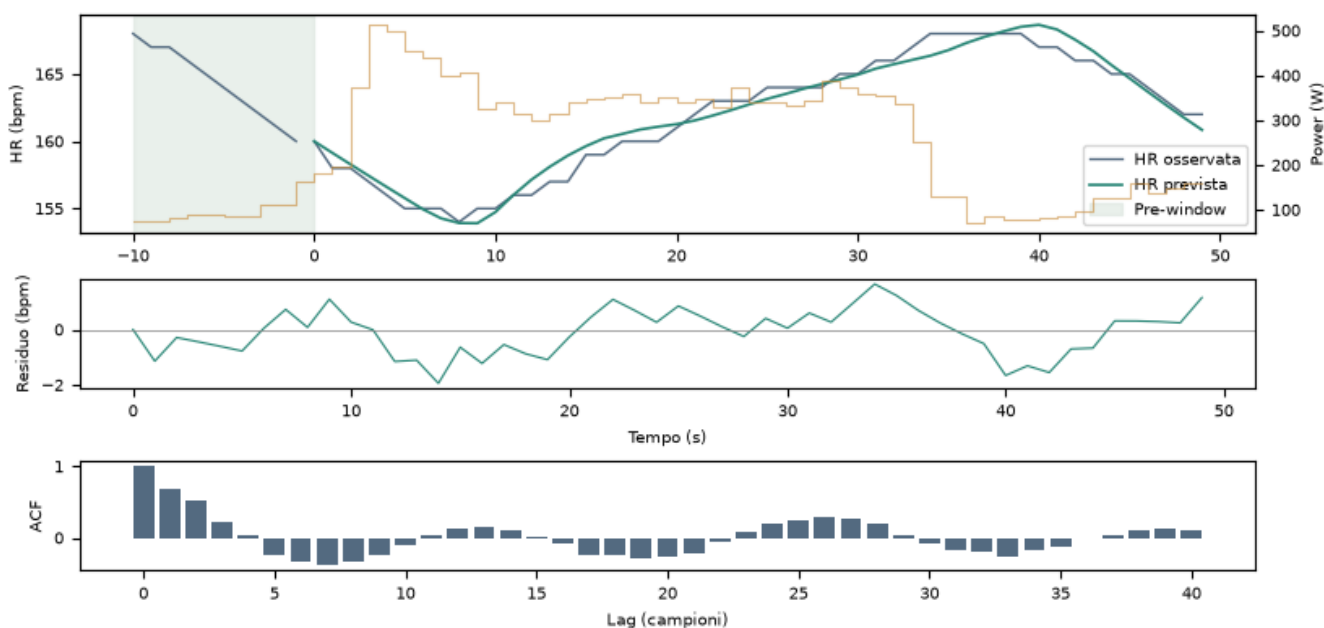
ID: 817afcbd82486c444de90a4d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=97 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.82246	0.66821	-0.087203	0.96525	1.0537
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.087203	0.82613	0.68284		-0.23126



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.59094	0.44424	-0.27977 / 1.4617	75.175	no	no
L	6.6832	0.33528	6.026 / 7.3404	5.0168	no	no
tau	115.81	87.682	-56.048 / 287.67	75.713	no	no
B	60	76.461	-89.863 / 209.86	127.43	si	si

rho=0.3654; Jacobian rank=4/4; cond=40058; accordo multistart=no.

Correlazioni: {"K_L": -0.8949360912797508, "K_tau": 0.9986421341498402, "K_B": -0.999919205909154, "L_tau": -0.8888347395751354, "L_B": 0.897818365818478, "tau_B": -0.9980913284375609}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 33.822 / 555.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 33.822; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_028-029.csv

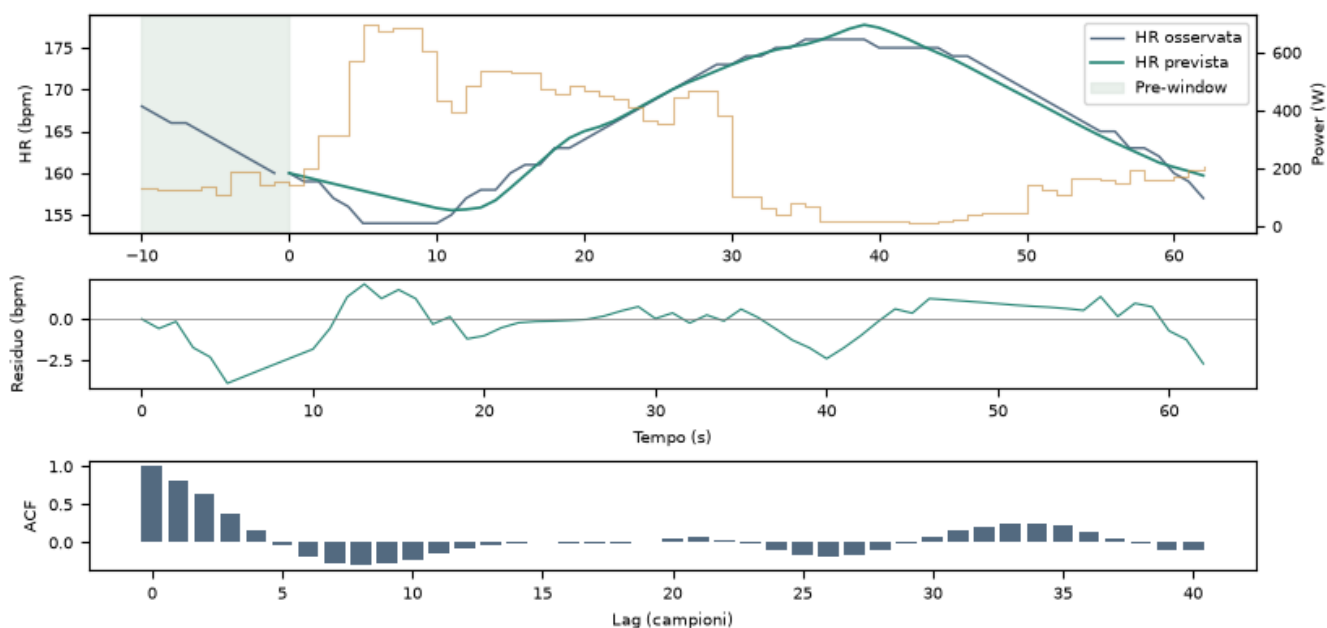
ID: 964197ddc49501314e2f4456 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=144 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.3462	1.011	-0.25857	0.96712	1.0084
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.25857	1.3317	0.82209	-0.028526	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.85712	1.0697	-1.2394 / 2.9537	124.8	no	no
L	9.2956	0.5197	8.277 / 10.314	5.5909	no	no
tau	234.26	292.34	-338.72 / 807.24	124.79	no	no
B	60	132.39	-199.49 / 319.49	220.65	si	si

rho=0.22498; Jacobian rank=4/4; cond=46970; accordo multistart=no.

Correlazioni: {"K_L": -0.9131088702143279, "K_tau": 0.9997199362739148, "K_B": -0.9999058201114297, "L_tau": -0.915835206112435, "L_B": 0.9123791684415195, "tau_B": -0.9993944588340389}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 114.17 / 2258.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 114.17; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_084-085.csv

ID: 9a66bc064c74937bc5b9c2e9 | Stato: fitted

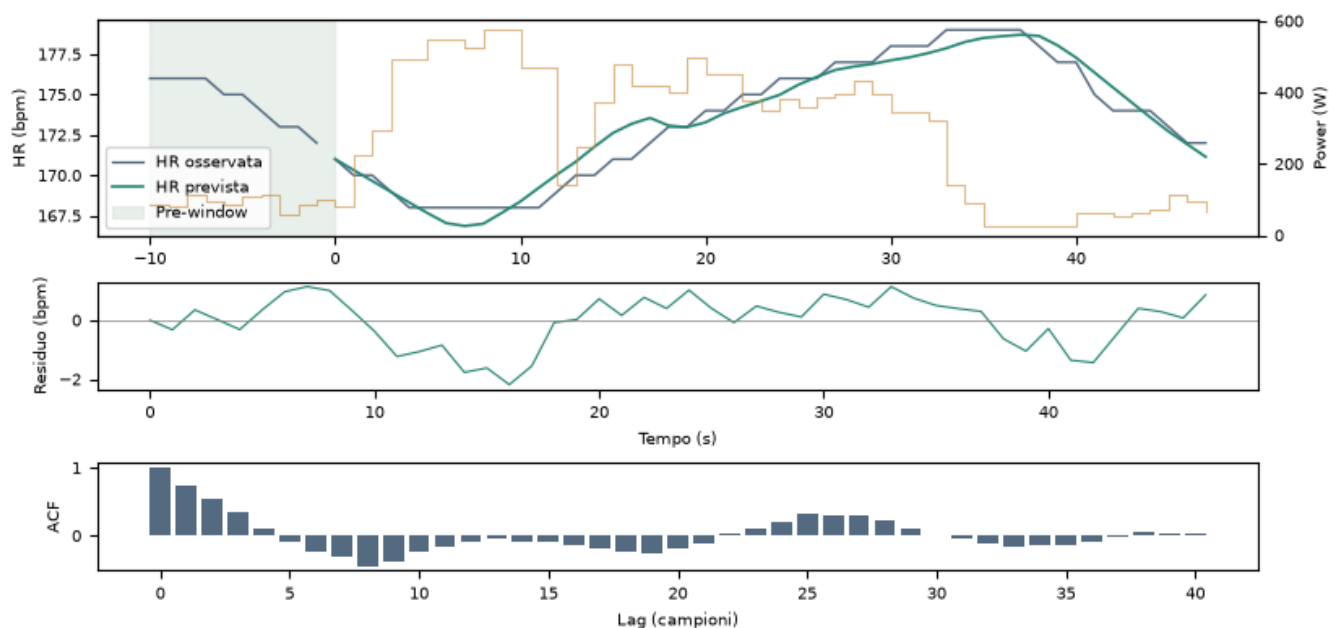
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=91.8 W; HR0=171 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=111 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.83599	0.66503	-0.031346	0.95051	1.0764

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.031346	0.84425	0.74358	-0.098185



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.54162	0.72908	-0.88738 / 1.9706	134.61	no	no
L	4.8267	0.39022	4.0618 / 5.5915	8.0847	no	no
tau	163.47	220.11	-267.94 / 594.88	134.64	no	no
B	60	152.56	-239.01 / 359.01	254.26	no	si

rho=0.25798; Jacobian rank=4/4; cond=79717; accordo multistart=no.

Correlazioni: {"K_L": -0.9031294467670188, "K_tau": 0.9993856738787376, "K_B": -0.9999583009977268, "L_tau": -0.9024683457944594, "L_B": 0.904344086626086, "tau_B": -0.9991092592827114}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 33.546 / 380.69

Griglia: 42441 coppie L/tau; SSE multistart originale: 33.546; selezione: multistart.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_002-003.csv

ID: 9c7bee85ac62ed78098de523 | Stato: fitted

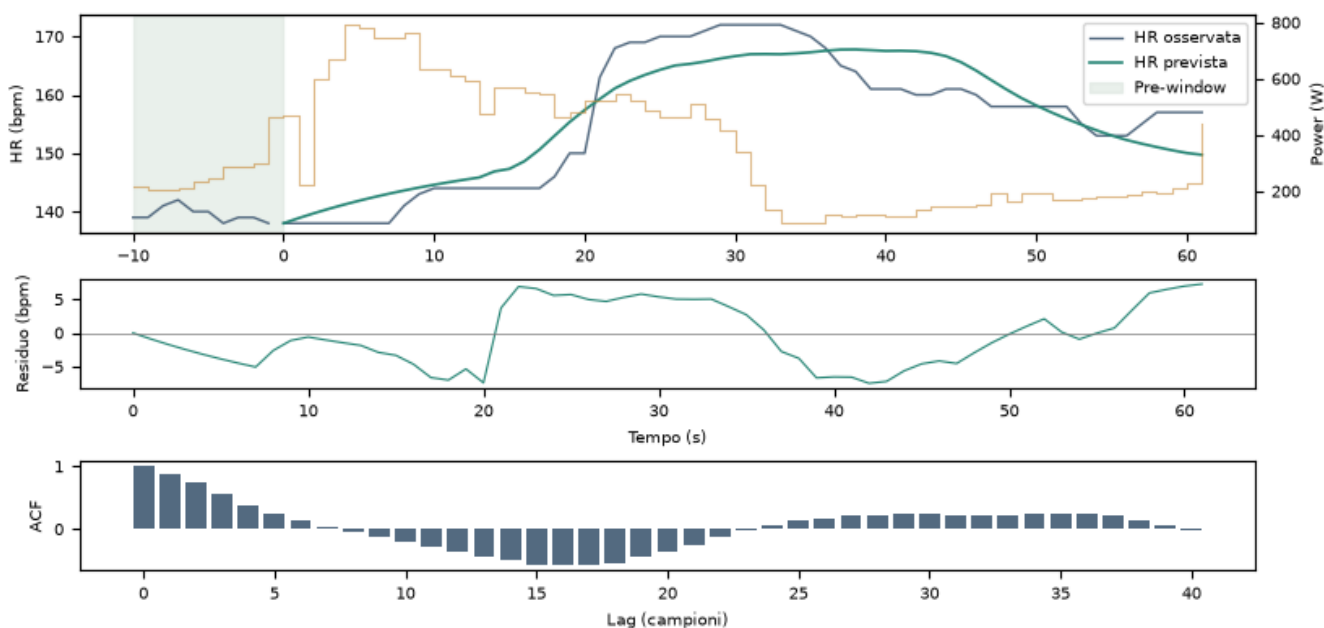
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=266.3 W; HR0=138 bpm.

HR(0) osservata fissata; B=151.39 bpm stimato; x(0)=-13.389 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
4.5338	3.9292	-0.357	0.83812	0.87568

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.357	4.5567	0.88813	0.2385



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.072407	0.01221	0.048476 / 0.096339	16.863	no	no
L	13.31	1.2334	10.893 / 15.728	9.2667	no	no
tau	14.738	2.9873	8.8827 / 20.593	20.27	no	no
B	151.39	1.3835	148.68 / 154.1	0.91385	no	no

rho=3.2359; Jacobian rank=4/4; cond=806.59; accordo multistart=no.

Correlazioni: {"K_L": -0.6017082601092764, "K_tau": 0.8183024404115065, "K_B": -0.6837258474546946, "L_tau": -0.8187591325668436, "L_B": 0.09594599257725656, "tau_B": -0.2779123776768696}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1274.5 / 3714.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 1324.7; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_010-011.csv

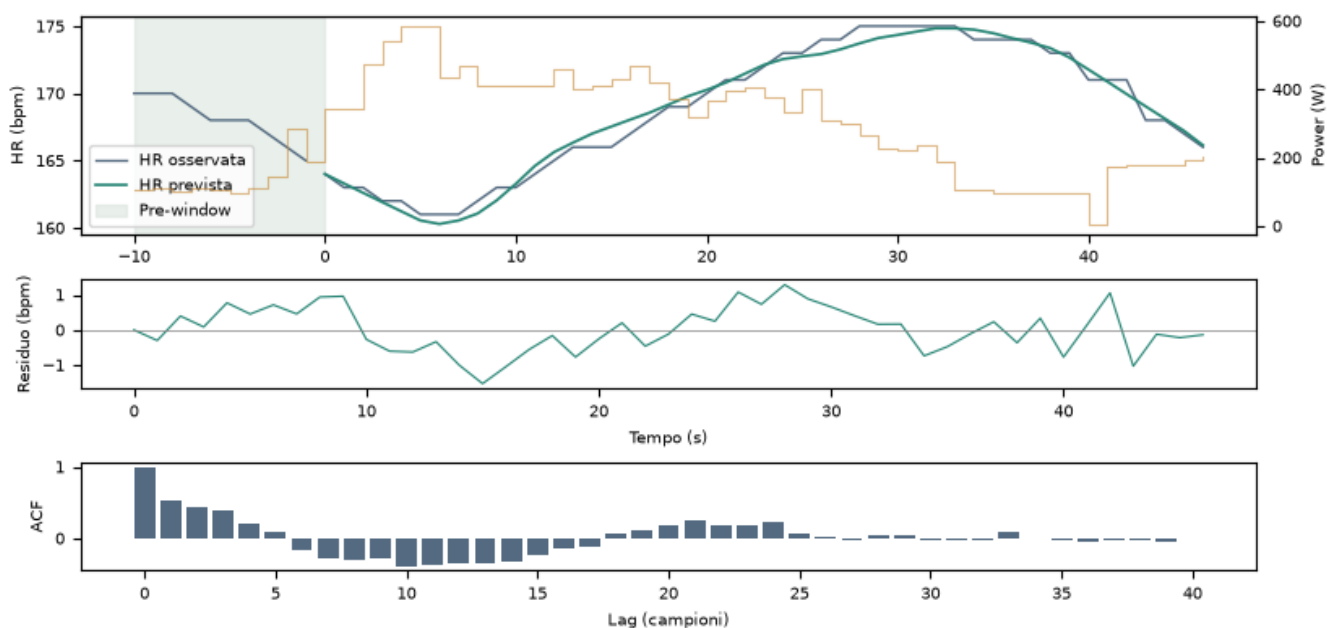
ID: 9d8b0d46644044c3e684759b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=135.8 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.6392	0.52757	0.016778	0.9818	1.0392
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.016778	0.64589	0.53193	0.093806	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.6716	0.61331	-0.53048 / 1.8737	91.32	no	no
L	5.5545	0.32882	4.9101 / 6.199	5.9198	no	no
tau	148.12	135.14	-116.75 / 412.99	91.238	no	no
B	60	99.326	-134.68 / 254.68	165.54	si	si

rho=0.27306; Jacobian rank=4/4; cond=58671; accordo multistart=no.

Correlazioni: {"K_L": -0.9350257813120031, "K_tau": 0.9994313837587985, "K_B": -0.9999317199899055, "L_tau": -0.937391651315833, "L_B": 0.935079678980622, "tau_B": -0.999040050238114}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 19.203 / 475.53

Griglia: 42441 coppie L/tau; SSE multistart originale: 19.203; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_086-087.csv

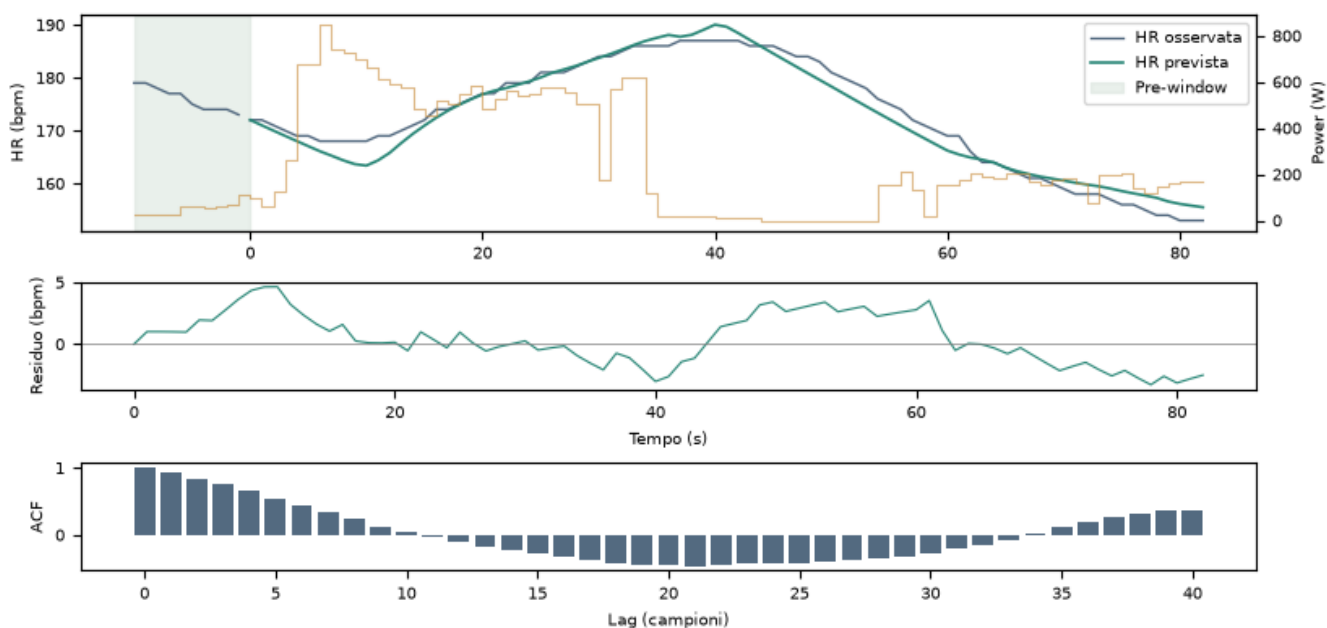
ID: a1ec45de9bf97f324b4258c9 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=53.1 W; HR0=172 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=112 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0861	1.6959	0.42242	0.96008	1.0151
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.42242	2.0553	0.92325	0.54291	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41863	0.17949	0.066821 / 0.77043	42.877	no	no
L	6.2826	0.35345	5.5898 / 6.9753	5.6258	no	no
tau	110.72	49.811	13.092 / 208.35	44.988	no	no
B	60	49.729	-37.47 / 157.47	82.882	si	si

rho=0.68386; Jacobian rank=4/4; cond=26683; accordo multistart=no.

Correlazioni: {"K_L": -0.7662865457750175, "K_tau": 0.998432941321063, "K_B": -0.9998839782064415, "L_tau": -0.7767005118098987, "L_B": 0.7688987503072647, "tau_B": -0.9985627023541782}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 361.2 / 8852.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 361.2; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_004-005.csv

ID: a7b44580dd83fe773f75f1f6 | Stato: fitted

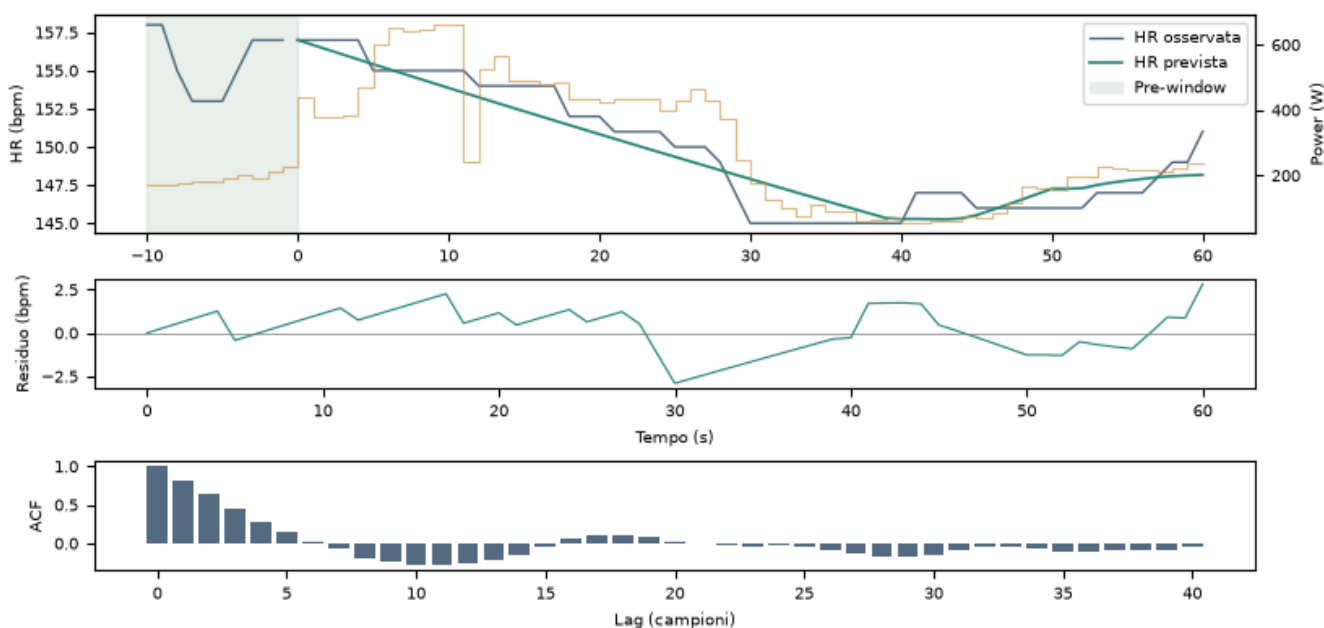
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=190.4 W; HR0=157 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=97 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.255	1.0571	0.18826	0.90587	0.9788

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.18826	1.2511	0.81067	0.15134



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41857	0.95209	-1.4475 / 2.2847	227.46	no	no
L	39.367	1.3672	36.687 / 42.047	3.4731	no	no
tau	304.55	660.8	-990.62 / 1599.7	216.98	no	no
B	60	200.05	-332.09 / 452.09	333.41	si	si

rho=0.067748; Jacobian rank=4/4; cond=30418; accordo multistart=no.

Correlazioni: {"K_L": -0.4987780512008351, "K_tau": 0.9991288414299151, "K_B": -0.999255276526424, "L_tau": -0.5160428035230931, "L_B": 0.5212661916590638, "tau_B": -0.9999080270472162}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 96.082 / 214.82

Griglia: 42441 coppie L/tau; SSE multistart originale: 96.082; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_024-025.csv

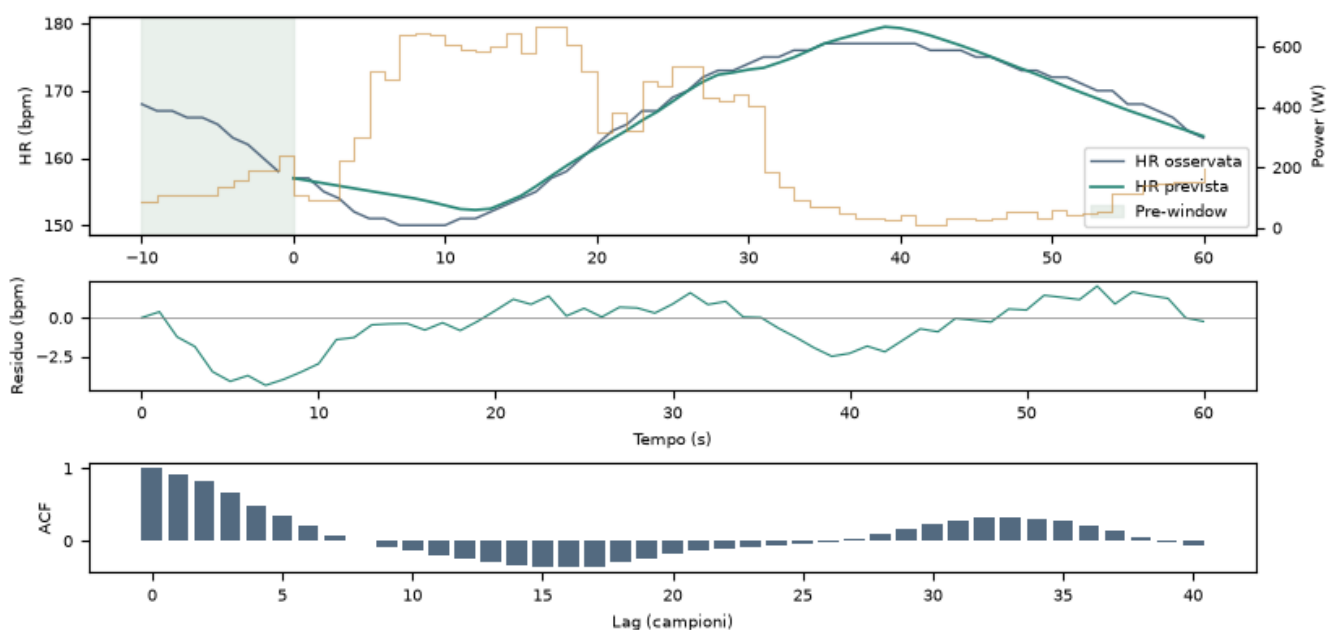
ID: c162ca94114fd93118d6d723 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=143.1 W; HR0=157 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=97 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6524	1.2326	-0.46738	0.97018	1.0079
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.46738	1.5981	0.90784	0.34909	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.96371	1.1801	-1.3494 / 3.2768	122.46	no	no
L	8.1504	0.56768	7.0378 / 9.2631	6.965	no	no
tau	253.38	310.65	-355.5 / 862.26	122.6	no	no
B	60	128.71	-192.27 / 312.27	214.52	si	si

rho=0.20463; Jacobian rank=4/4; cond=38656; accordo multistart=no.

Correlazioni: {"K_L": -0.9110596500041687, "K_tau": 0.999590648433662, "K_B": -0.9996743809124705, "L_tau": -0.9137806075105863, "L_B": 0.90960541529196, "tau_B": -0.9986516763493415}

Multistart: 16 tentativi, 14 convergenti; SSE min/max: 166.57 / 2595.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 166.57; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_022-023.csv

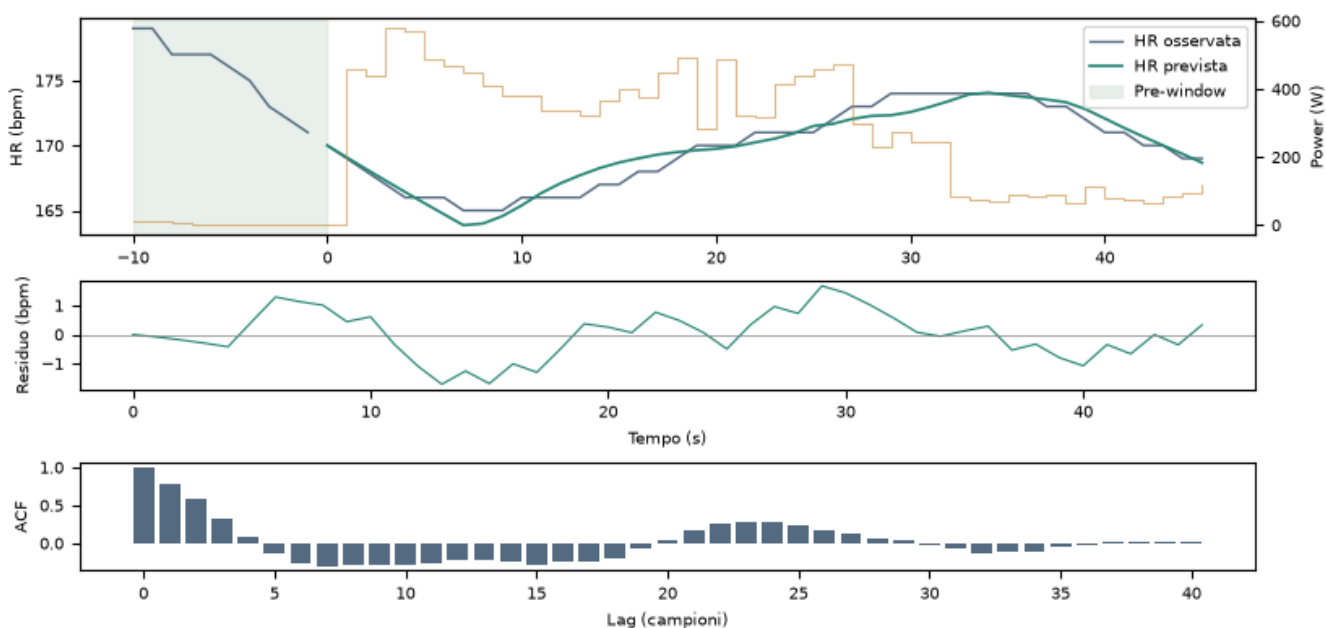
ID: c2b578d2d44afb5937d7392e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=2.9 W; HR0=170 bpm.

HR(0) osservata fissata; B=108.12 bpm stimato; x(0)=61.876 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.79083	0.63029	-0.0013544	0.92925	1.1326
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.0013544	0.79957	0.77822	-0.12423	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21304	0.15202	-0.084924 / 0.511	71.359	no	no
L	6.3352	0.38112	5.5882 / 7.0822	6.016	no	no
tau	67.143	48.388	-27.697 / 161.98	72.066	no	no
B	108.12	44.088	21.711 / 194.54	40.776	no	no

rho=0.57586; Jacobian rank=4/4; cond=50323; accordo multistart=no.

Correlazioni: {"K_L": -0.8793105827166371, "K_tau": 0.9977127023120576, "K_B": -0.9999214640433834, "L_tau": -0.8683326656459327, "L_B": 0.8829573528528912, "tau_B": -0.9972393452331737}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 28.769 / 370.62

Griglia: 42441 coppie L/tau; SSE multistart originale: 28.769; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_008-009.csv

ID: c9f23a286f8476bb5c88043f | Stato: fitted

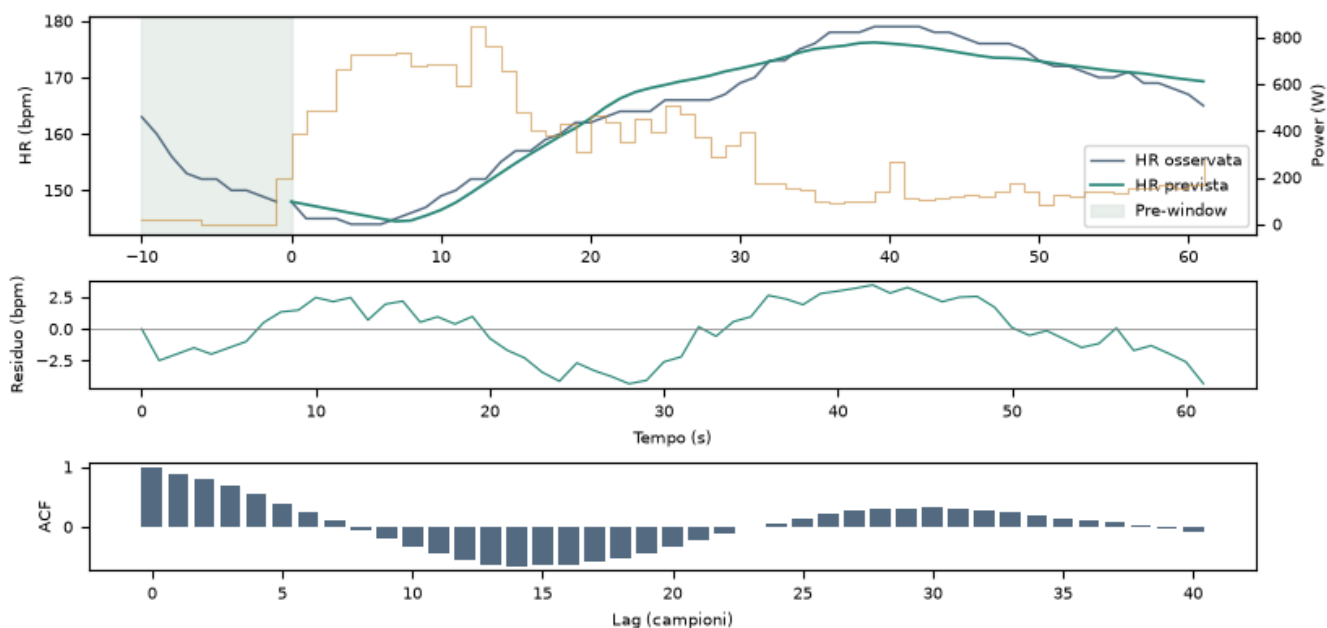
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=28.8 W; HR0=148 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=88 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.2336	1.9219	-0.079383	0.96128	0.90602

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.079383	2.2504	0.87777	0.39924



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.56702	0.60001	-0.609 / 1.7431	105.82	no	no
L	7.4499	0.77267	5.9355 / 8.9644	10.372	no	no
tau	172.61	175.63	-171.63 / 516.84	101.75	no	no
B	60	113.96	-163.37 / 283.37	189.94	si	si

rho=0.31024; Jacobian rank=4/4; cond=44745; accordo multistart=no.

Correlazioni: {"K_L": -0.920081389501601, "K_tau": 0.9987550947659674, "K_B": -0.9990956828072067, "L_tau": -0.9243155022426193, "L_B": 0.9176401735345575, "tau_B": -0.9958853571931232}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 309.31 / 1849.7

Griglia: 42441 coppie L/tau; SSE multistart originale: 309.31; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260407082440__laps_057-058.csv

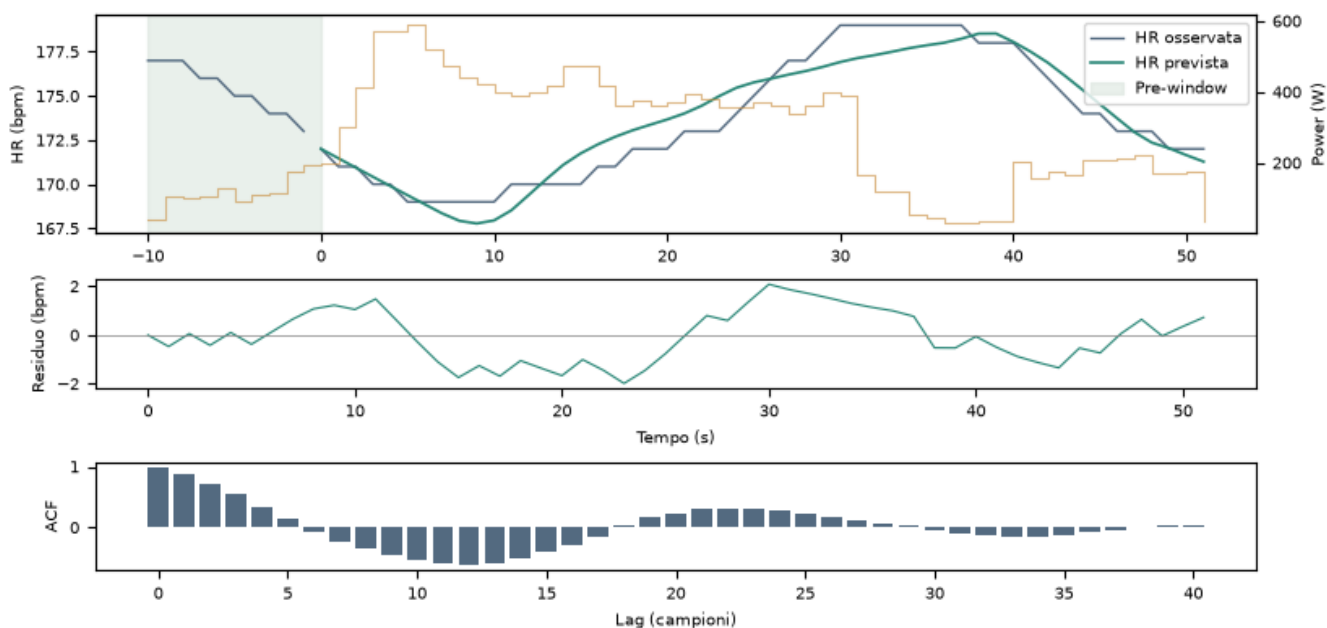
ID: d92f7b64a4dde6f36f04c793 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=117.1 W; HR0=172 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=112 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0591	0.89567	-0.040246	0.90683	1.0754
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.040246	1.0687	0.86752	0.14189	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.64248	1.5294	-2.3551 / 3.64	238.04	no	no
L	7.6228	0.64111	6.3662 / 8.8794	8.4104	no	no
tau	211.1	502.42	-773.64 / 1195.8	238	no	no
B	60	270.62	-470.41 / 590.41	451.03	no	si

rho=0.20548; Jacobian rank=4/4; cond=99156; accordo multistart=si.

Correlazioni: {"K_L": -0.917049652820725, "K_tau": 0.9997174967452753, "K_B": -0.9999802812793973, "L_tau": -0.9168280764104221, "L_B": 0.9179440317879087, "tau_B": -0.9996066351896484}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 58.328 / 58.328

Griglia: 42441 coppie L/tau; SSE multistart originale: 58.328; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_006-007.csv

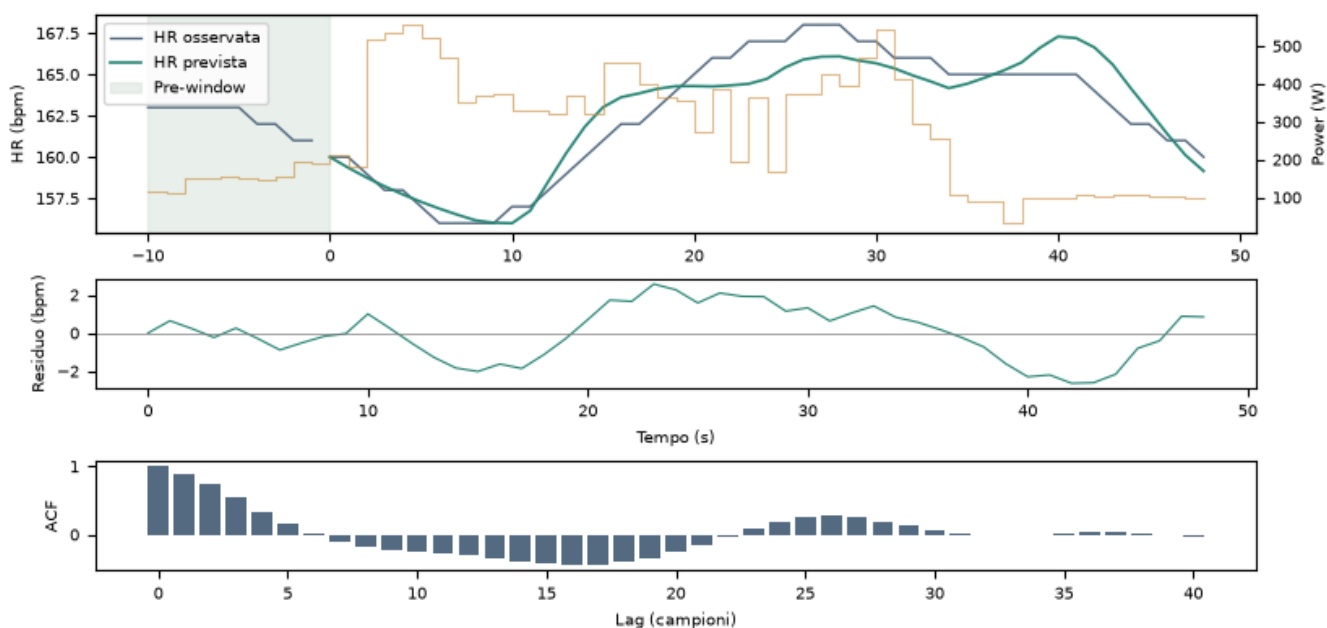
ID: dad73a96a82b268da0cc6e01 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=152.3 W; HR0=160 bpm.

HR(0) osservata fissata; B=152.62 bpm stimato; x(0)=7.375 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.381	1.1443	-0.010218	0.86496	0.94135
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.010218	1.3953	0.89996		0.15335



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.065657	0.0091048	0.047811 / 0.083502	13.867	no	no
L	8.5507	0.36597	7.8334 / 9.268	4.28	no	no
tau	10.926	1.9509	7.1023 / 14.75	17.856	no	no
B	152.62	1.4283	149.83 / 155.42	0.93582	no	no

rho=3.6106; Jacobian rank=4/4; cond=1662.6; accordo multistart=no.

Correlazioni: {"K_L": -0.6540127989771949, "K_tau": 0.8515025522366761, "K_B": -0.9808233486928031, "L_tau": -0.717332255411794, "L_B": 0.6270617335002809, "tau_B": -0.7934562026426106}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 93.449 / 451.79

Griglia: 42441 coppie L/tau; SSE multistart originale: 98.196; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_006-007.csv

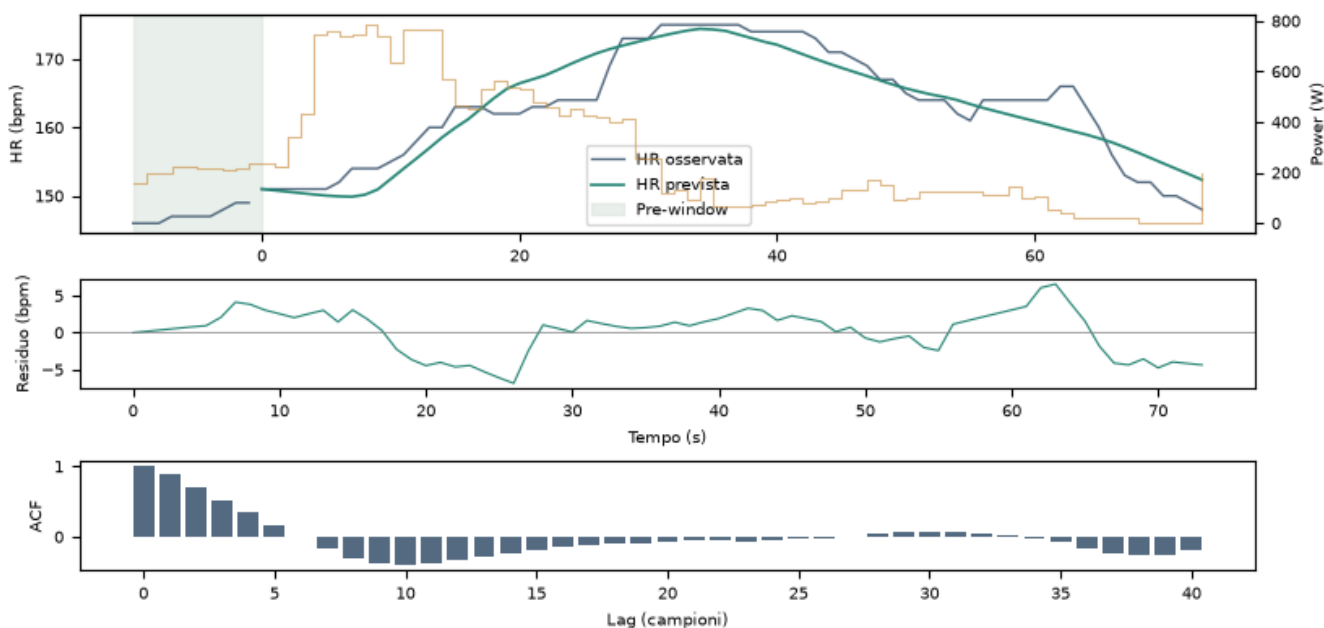
ID: e4d5945efc8258a4c905f544 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=210 W; HR0=151 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=91 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.9273	2.4089	0.17329	0.87005	0.90891
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.17329	2.9421	0.88302	0.1636	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.4687	6.4216	-11.118 / 14.055	437.23	no	no
L	4.7295	1.0357	2.6995 / 6.7595	21.899	no	no
tau	464.42	2025.3	-3505.1 / 4434	436.09	no	no
B	60	452.48	-826.86 / 946.86	754.14	si	si

rho=0.147; Jacobian rank=4/4; cond=84281; accordo multistart=no.

Correlazioni: {"K_L": -0.873369202523552, "K_tau": 0.9999508562284989, "K_B": -0.9998905727304964, "L_tau": -0.8755578960534033, "L_B": 0.8701257284056463, "tau_B": -0.9997256741160213}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 634.1 / 3482.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 634.1; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_020-021.csv

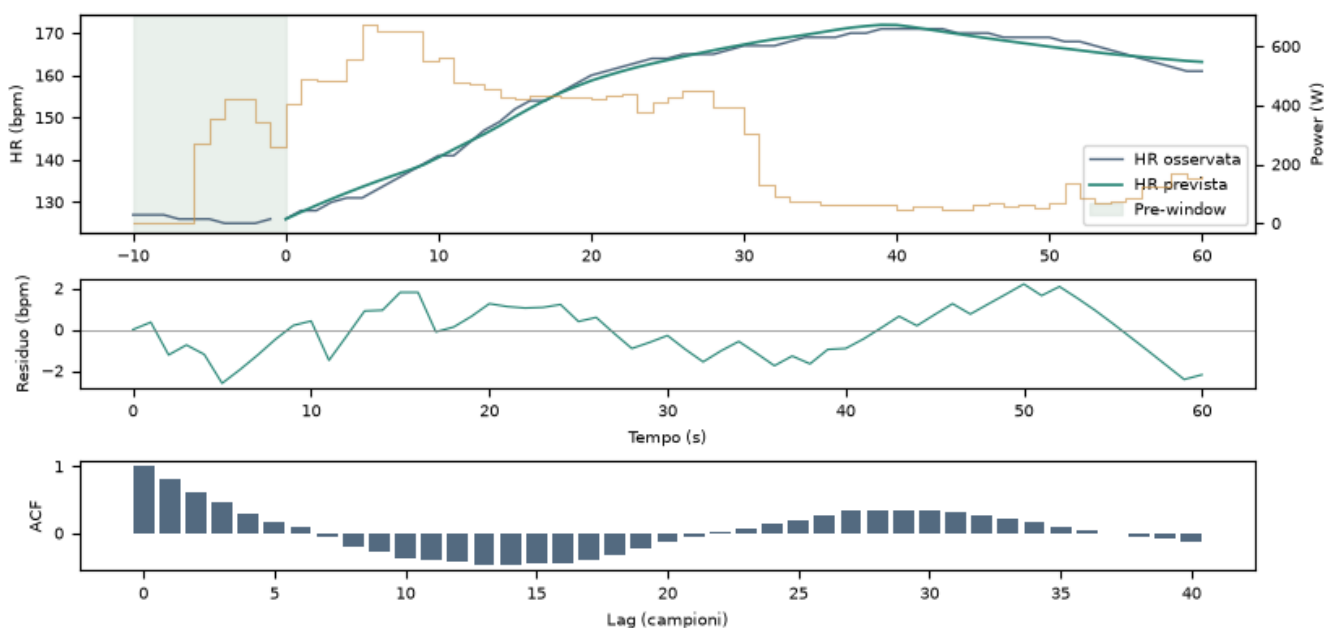
ID: e8702c0b003c461e22e469f6 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=207.8 W; HR0=126 bpm.

HR(0) osservata fissata; B=167.02 bpm stimato; x(0)=-41.018 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2045	1.0245	-0.071779	0.9924	1.0214
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.071779	1.2123	0.80693	0.17027	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.074332	0.0064786	0.061634 / 0.08703	8.7157	no	no
L	8.7582	0.60771	7.5671 / 9.9493	6.9387	no	no
tau	24.391	1.5665	21.321 / 27.461	6.4224	no	no
B	167.02	0.72812	165.59 / 168.45	0.43595	no	no

rho=2.1008; Jacobian rank=4/4; cond=1226.6; accordo multistart=no.

Correlazioni: {"K_L": -0.7035690077352864, "K_tau": 0.8996023643399765, "K_B": 0.4295671228774496, "L_tau": -0.8636410322320284, "L_B": -0.7814923655504001, "tau_B": 0.7530954492778077}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 88.494 / 964.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 88.494; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_035-036.csv

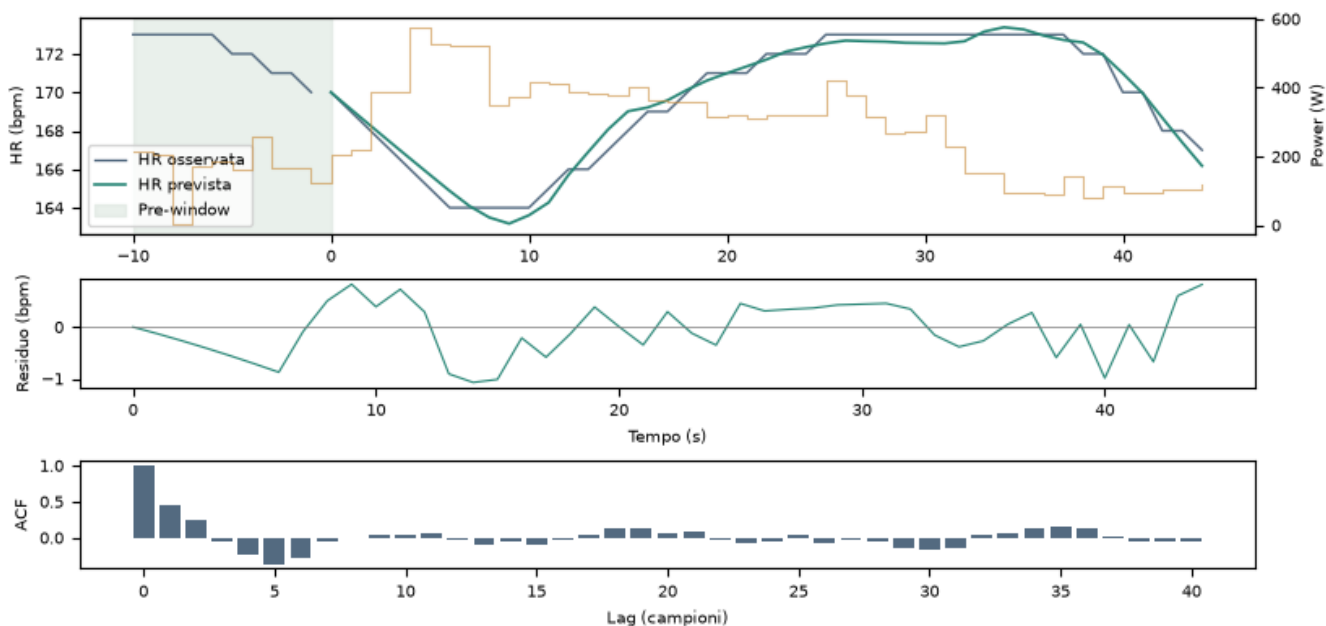
ID: f0184f3a8499a75fb100c8df | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=166.5 W; HR0=170 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=110 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.51228	0.42722	-0.052949	0.97405	1.1343
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.052949	0.51529	0.45661	-0.36138	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.71627	0.61123	-0.48173 / 1.9143	85.335	no	no
L	6.783	0.212	6.3675 / 7.1985	3.1255	no	no
tau	125.7	107.77	-85.525 / 336.92	85.735	no	no
B	60	93.498	-123.26 / 243.26	155.83	si	si

rho=0.29609; Jacobian rank=4/4; cond=51009; accordo multistart=no.

Correlazioni: {"K_L": -0.889404253128163, "K_tau": 0.9995195306358922, "K_B": -0.9999836577427071, "L_tau": -0.886458077395279, "L_B": 0.8907090968359367, "tau_B": -0.999450759240328}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11.809 / 447.25

Griglia: 42441 coppie L/tau; SSE multistart originale: 11.809; selezione: profile_grid.

Segnalazioni: at_bounds, multistart_disagreement

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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